TOKYO METROPOLITAN UNIVERSITY
東京都立大学

Doctoral Dissertation Defense

学位（博士）論文公聴会

A Study on Badminton Video Analysis Based on Deep Learning Models

（ 深層学習モデルに基づくバドミントン映像解析の研究 ）

Graduate School of Systems Design, Department of Electronics and Information System Engineering

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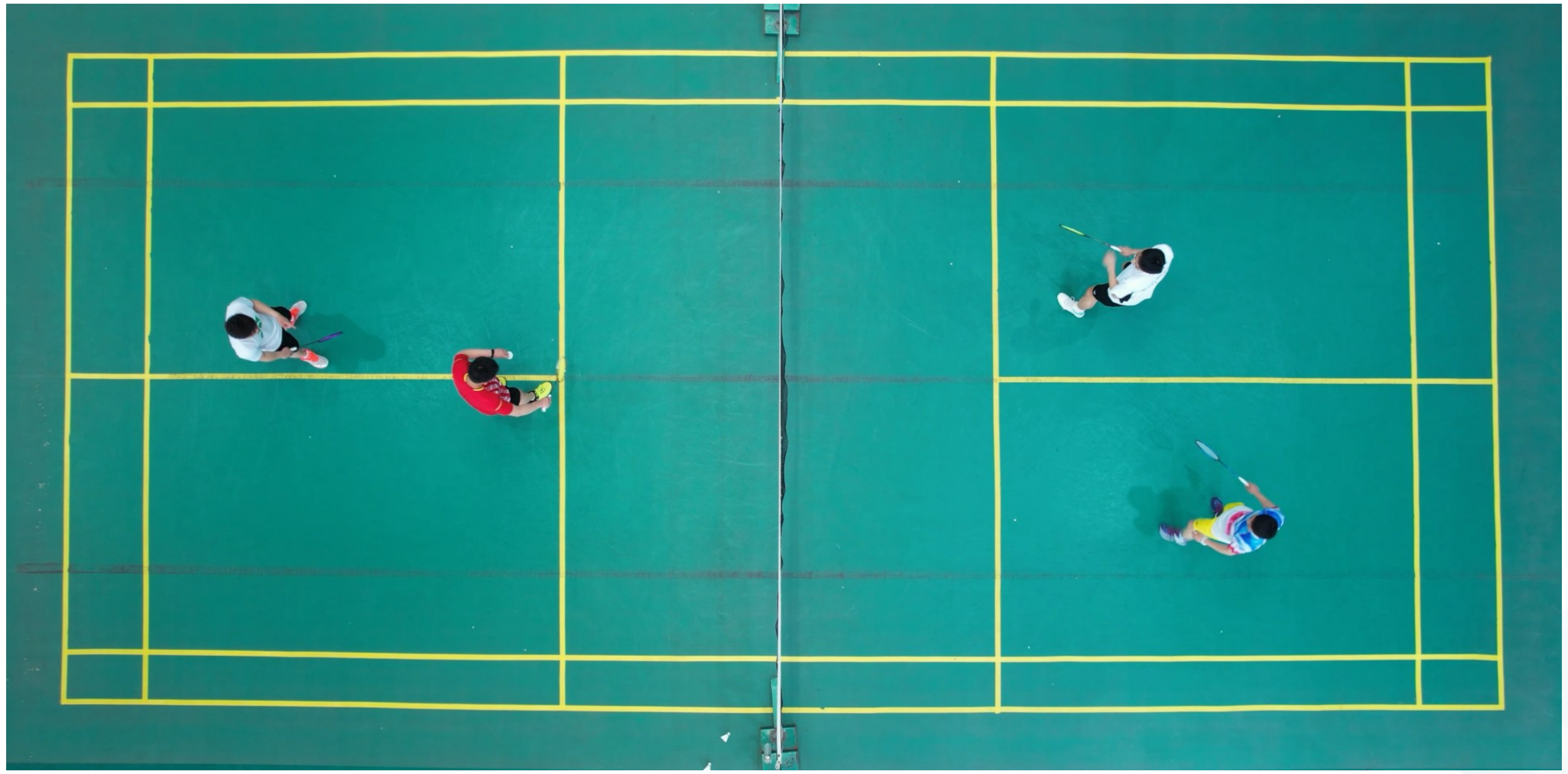
Outlines

- Chapter 1: Introduction
- Chapter 2: Literature Review
- Chapter 3: Player Detection
- Chapter 4: Shuttlecock Detection
- Chapter 5: Racket Detection
- Chapter 6: Dominant Area
- Chapter 7: Conclusions and Future Works

Introduction

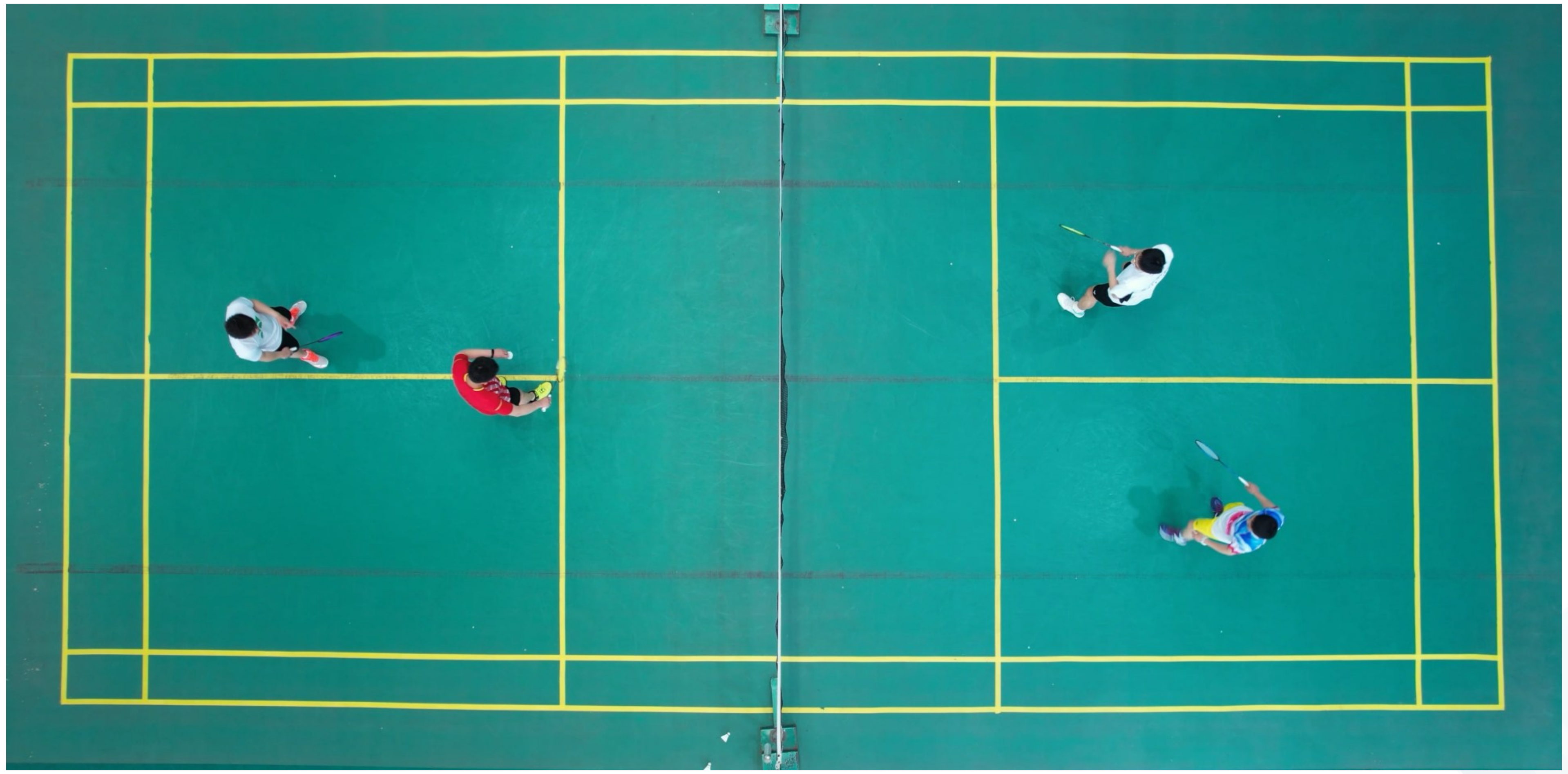
Background



- Badminton is a fast racket sport with rapid player movement, high-speed shuttlecock motion, and frequent interactions among players, rackets, and the shuttlecock.
- Tactical understanding, especially in doubles matches, depends on spatial positioning, court coverage, and player coordination.

Introduction

Background (Cont.)



- Although computer vision has advanced sports video analysis, badminton remains less explored than other sports.

- Challenges such as the shuttlecock's small size, high speed, racket rotation, occlusion, and complex interactions make analysis difficult, and many existing methods focus only on isolated detections rather than overall spatial dynamics.

Introduction

Motivation and Study Objectives

Most existing badminton analysis focuses on individual detections and does not explicitly model spatial control on the court, which is crucial for tactical understanding in doubles play.

The objectives of this study are:

- Develop a pipeline for detecting and tracking players, shuttlecock, and rackets.
- Incorporate oriented racket detection with spatially meaningful representations.
- Introduce a dominant area visualization that represents court control by integrating player movement, shuttlecock position, and racket information.

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Chapter 2

Literature Review

Player Detection using Faster R-CNN

Rahmad et al. (2019) introduced an automated badminton player detection method using Faster R-CNN on broadcast match videos. Players were detected from extracted image frames using bounding boxes

Figure 2.1. Example results of badminton player detection from broadcast match videos using Faster R-CNN

U-Net

U-Net is an encoder–decoder convolutional network originally developed for biomedical image segmentation (Falk et al., 2019). It is widely applied to tasks such as cell detection and segmentation in microscopy images. Skip connections between the encoder and decoder help preserve spatial details for accurate pixel-wise predictions.

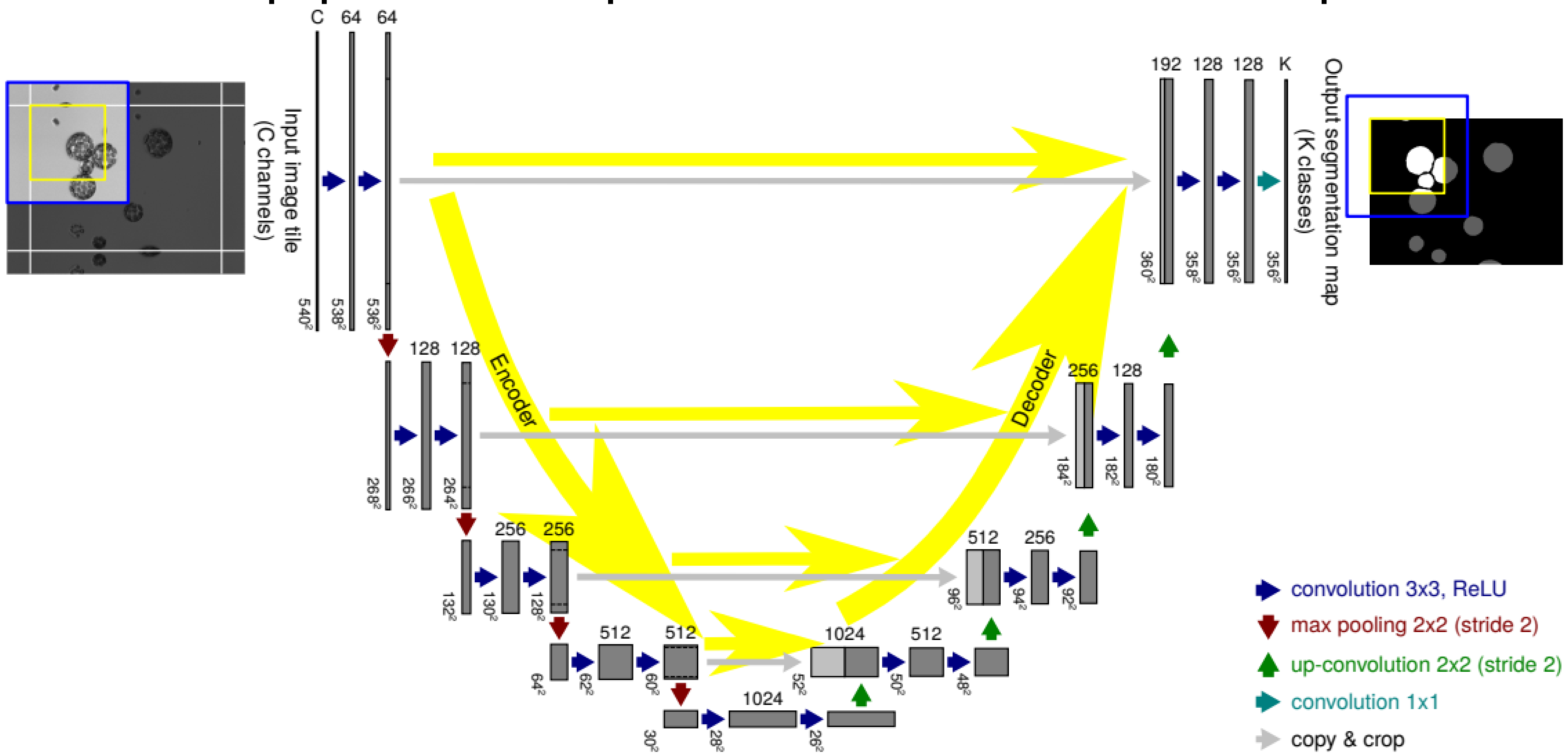

Figure 2.2. U-Net encoder–decoder architecture with skip connections for biomedical image segmentation.

TrackNetV2

TrackNetV2 was proposed to detect the shuttlecock from badminton broadcast match videos (Sun et al., 2020). It consists of a U-Net-based encoder-decoder backbone.

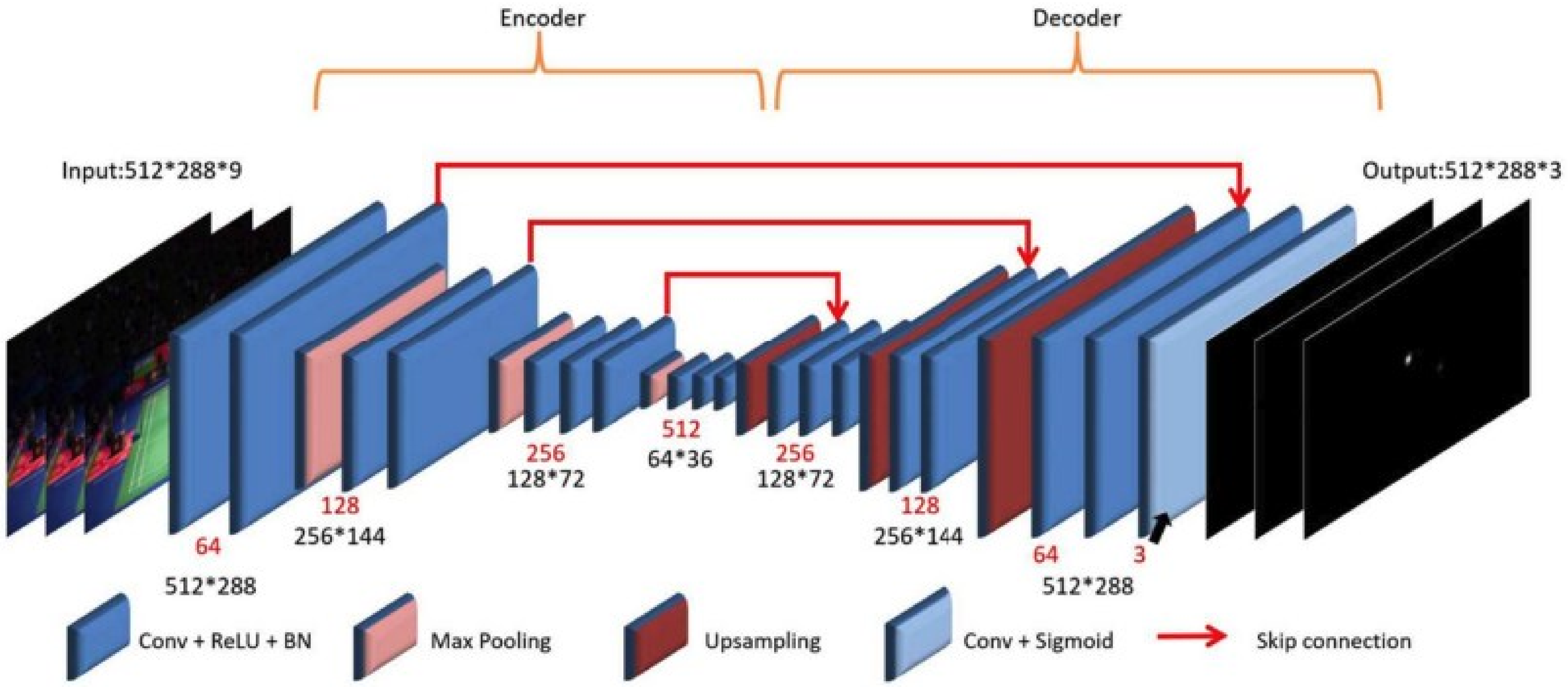

Figure 2.3. TrackNetV2 neural network architecture for detection of shuttlecocks

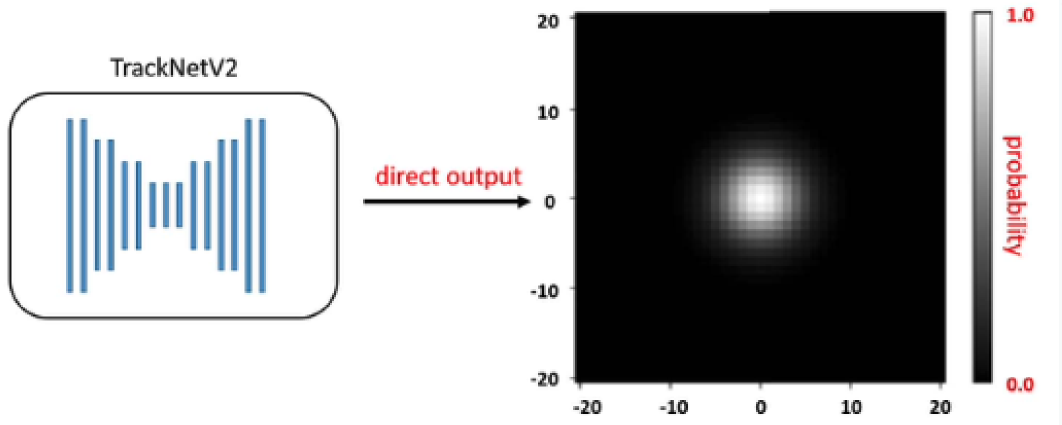

Figure 2.4. TrackNetV2 directly generates a real-valued array to have the heatmap.

Literature Review

VIRD

VIRD is Immersive VR based badminton match analysis system that integrates 3D reconstruction from video (Lin et al., 2023). It enables a top-down, interactive analysis workflow that supports tactical understanding, efficient match review, and effective communication between coaches and players.

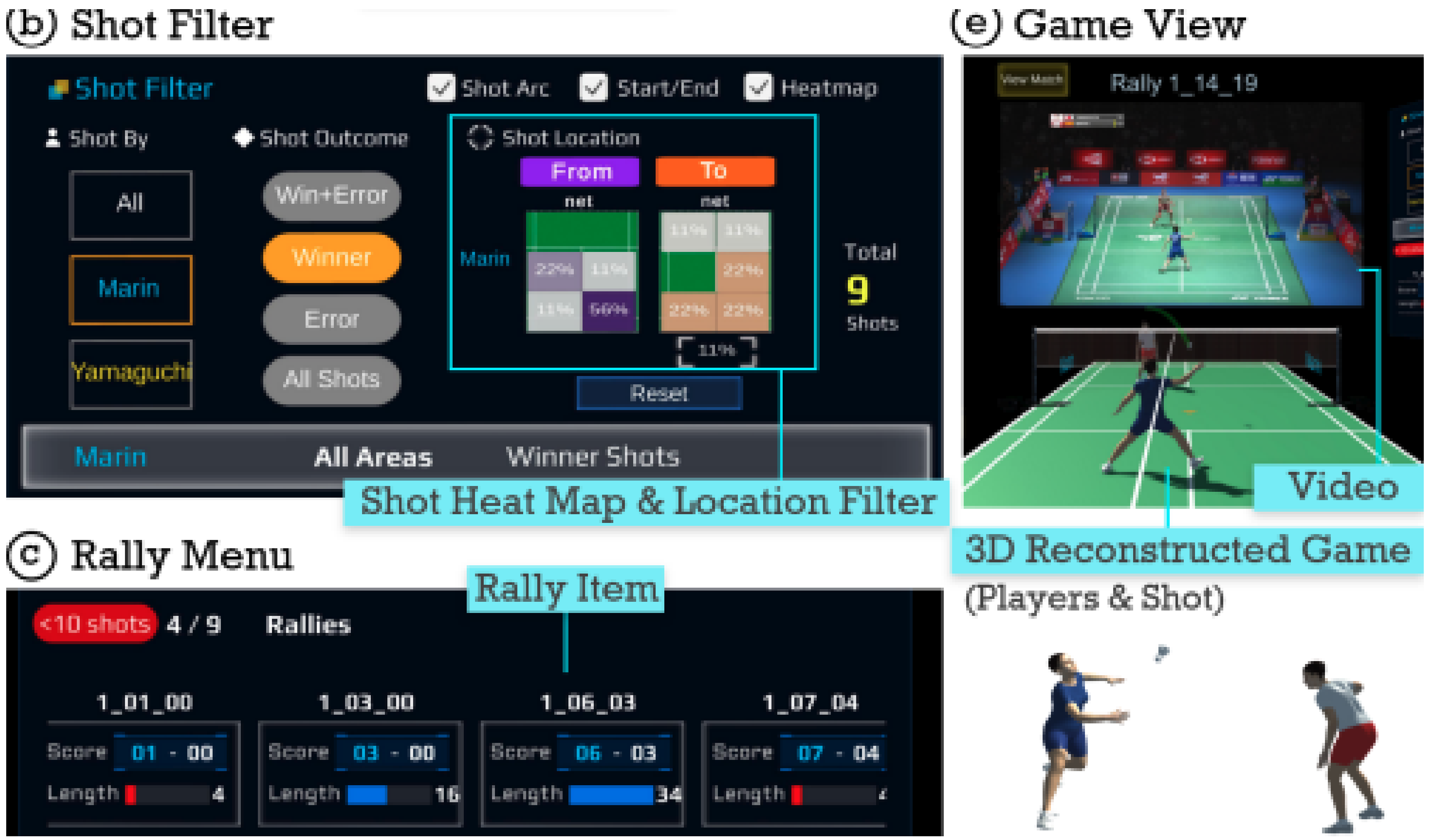

Figure 2.5. Display of VIRD with the match animation from badminton match

freely in the virtual court and watch the game from different angles. Further, all of them were excited seeing the moving bird and the 3D reconstructed game. Throughout the analysis, experts felt engaged and interactive, *"I like how I can move my body around and face the shot. I find that more beneficial than just looking at a TV screen or computer"* (C1). Experts also expressed their tendency to view a match in 3D and refer to the video only when the 3D view did not make sense. A coach also suggested adding rackets to improve the 3D view. Using VIRD, experts obtained deeper insights on the spatial aspects in their analysis. P1 observed most of his winners were hit downwards from his backhand side from analyzing the shot locations and trajectories.

Literature Review

Football player dominant region

Player control area analysis was originally developed in football to represent pitch regions that a player can reach before any other player (Caetano et al., 2021) . This concept provides a foundation for extending player area analysis to other sports, including badminton

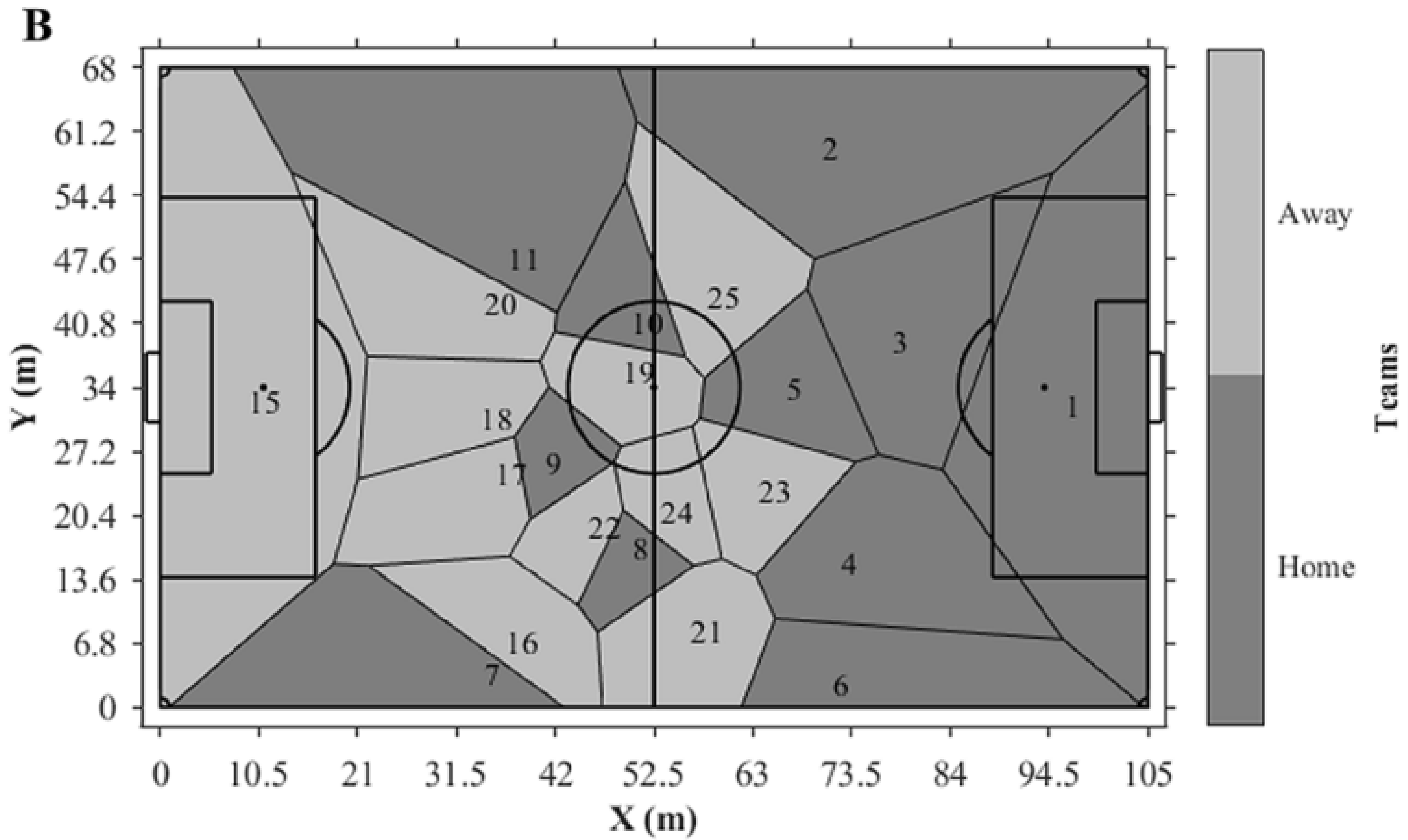

Figure 2.5 Example of player regions in football computed using Voronoi regions.

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Chapter 3

Player Detection

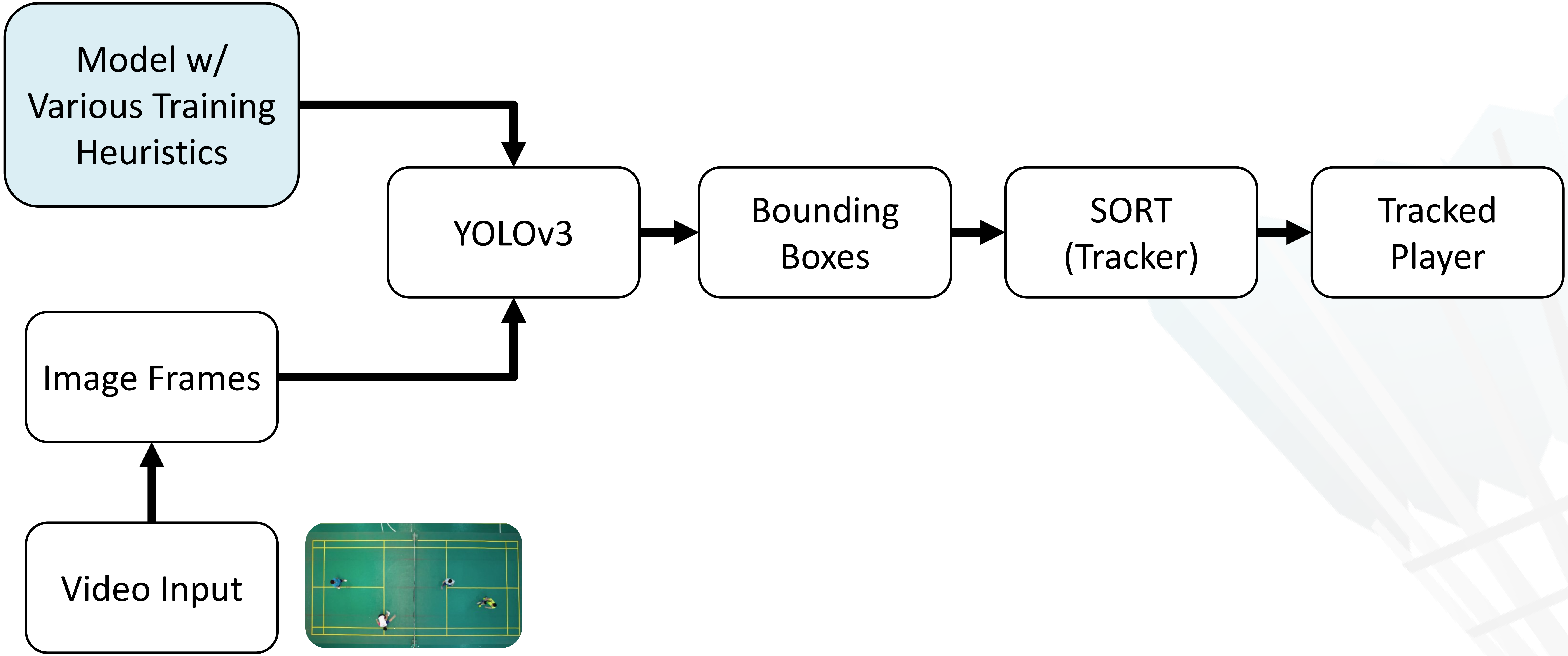
Introduction

- Accurate player detection in badminton videos remains challenging due to rapid player movements, frequent occlusions, and significant scale variations.
- These factors often lead to unstable training behavior and inconsistent detection performance when using standard training settings.
- To address this problem, this study investigates various training heuristics applied during model training.
- The objective is to improve training stability and detection robustness without modifying the underlying network architecture.

Chapter 3

Player Detection

Method : Pipeline



Player Detection

Method : Data Preprocessing and Augmentation

Apply Normalization, Cropping, Contrast Adjustment, HSV Augmentation to make the model more robust to different lighting, sizes, and positions.

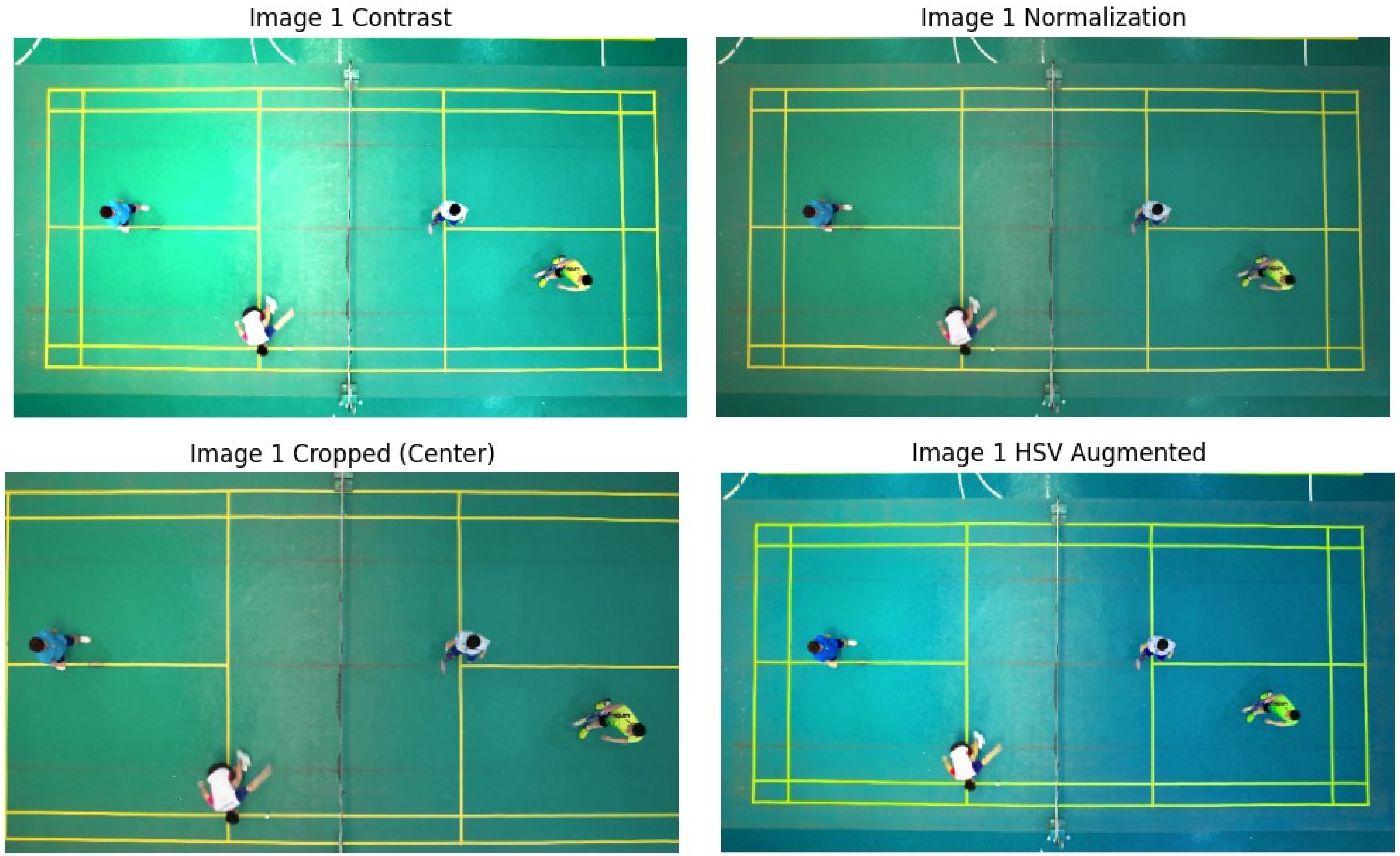


Figure 3.1. Data preprocessing and data augmentation used in player detection.

Player Detection

Method : CosineLR

Cosine Learning Rate Scheduler, adjust the learning rate during training with cosine scheduler to make learning more stable and accurate.

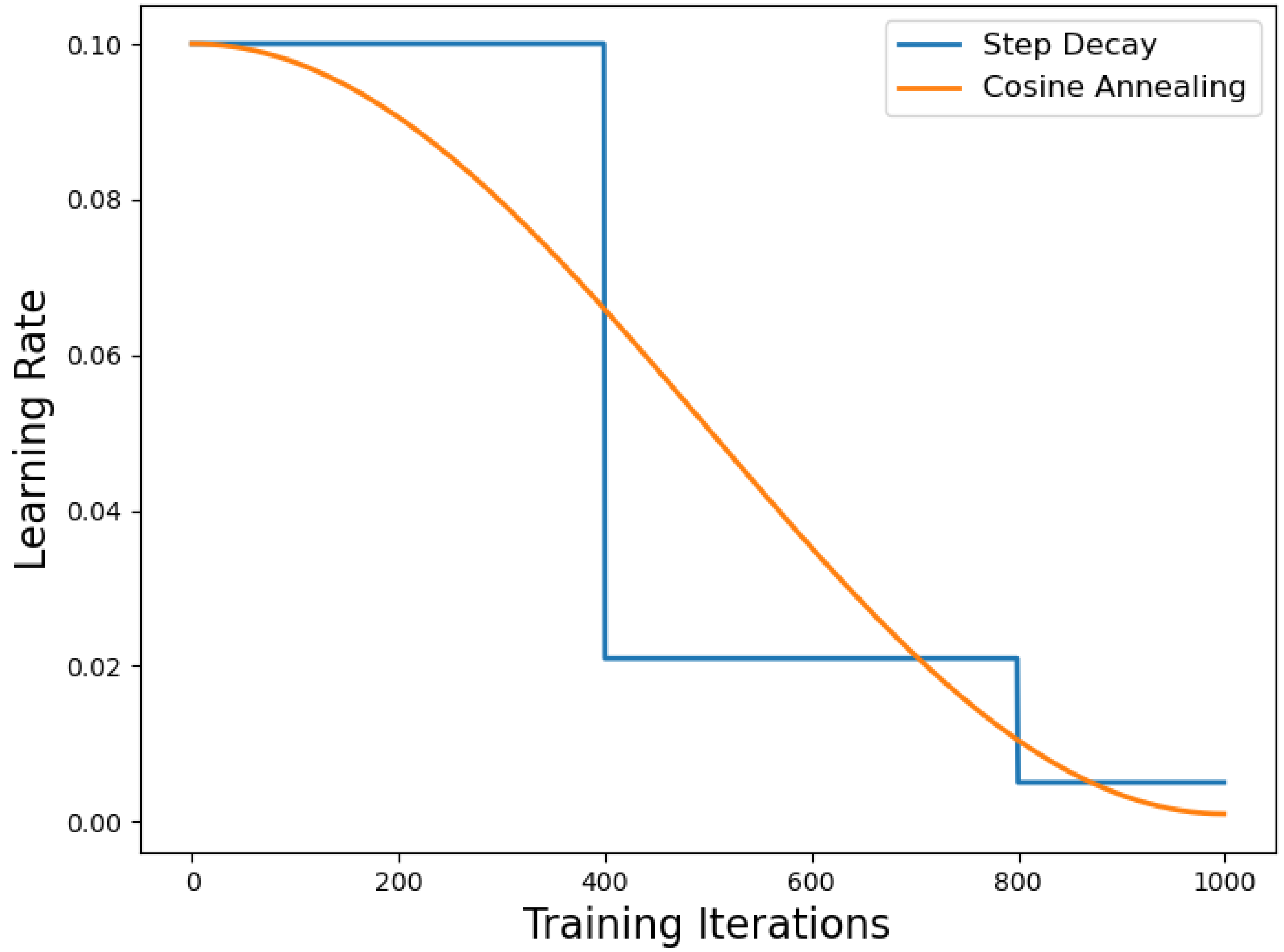


Figure 3.2. The difference between step decay and cosine learning rate.

Player Detection

Method : SyncBN

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Synchronized Batch Normalization (BN), share batch normalization statistics across GPUs to improve consistency and speed up training.

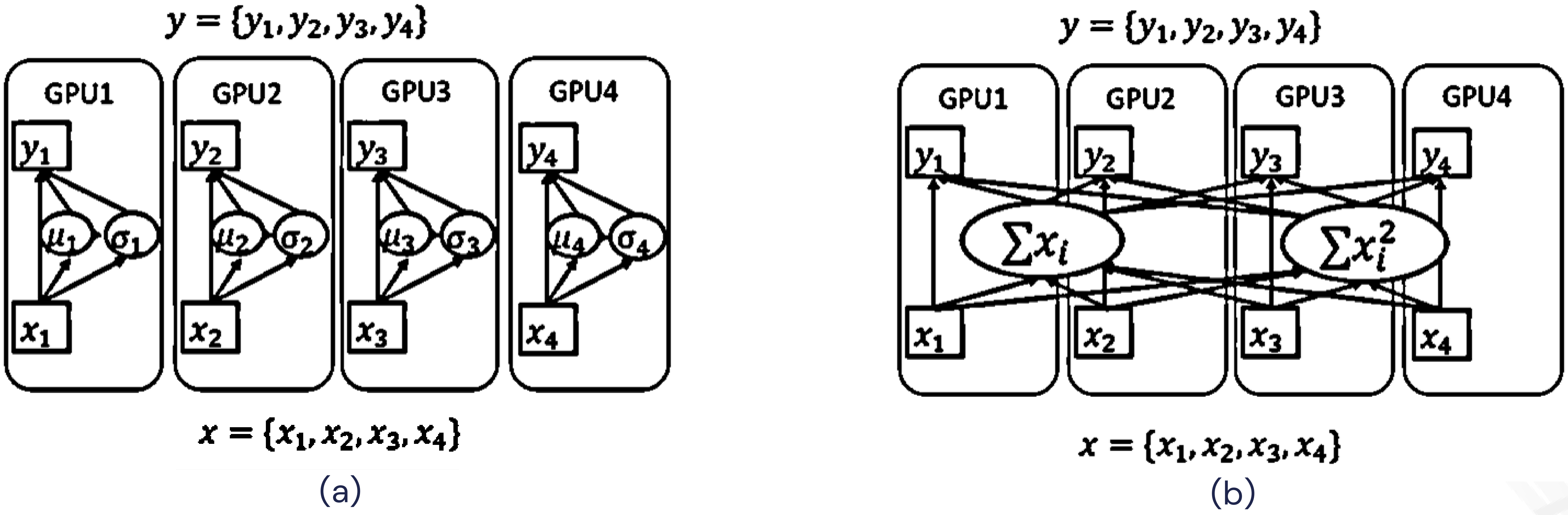


Figure 3.3. The difference between batch normalization (a) and synchronized batch normalization (b)

Player Detection

Method : Image Mixup

Combine two images into one by blending their pixels. This makes the training data more diverse and improves detection accuracy.

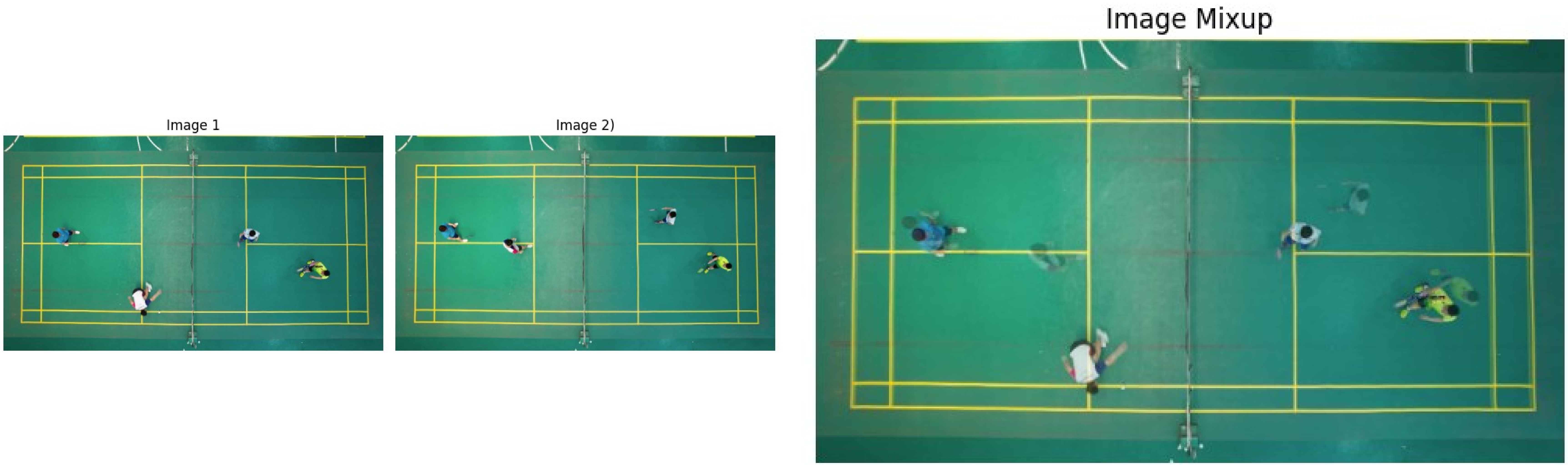


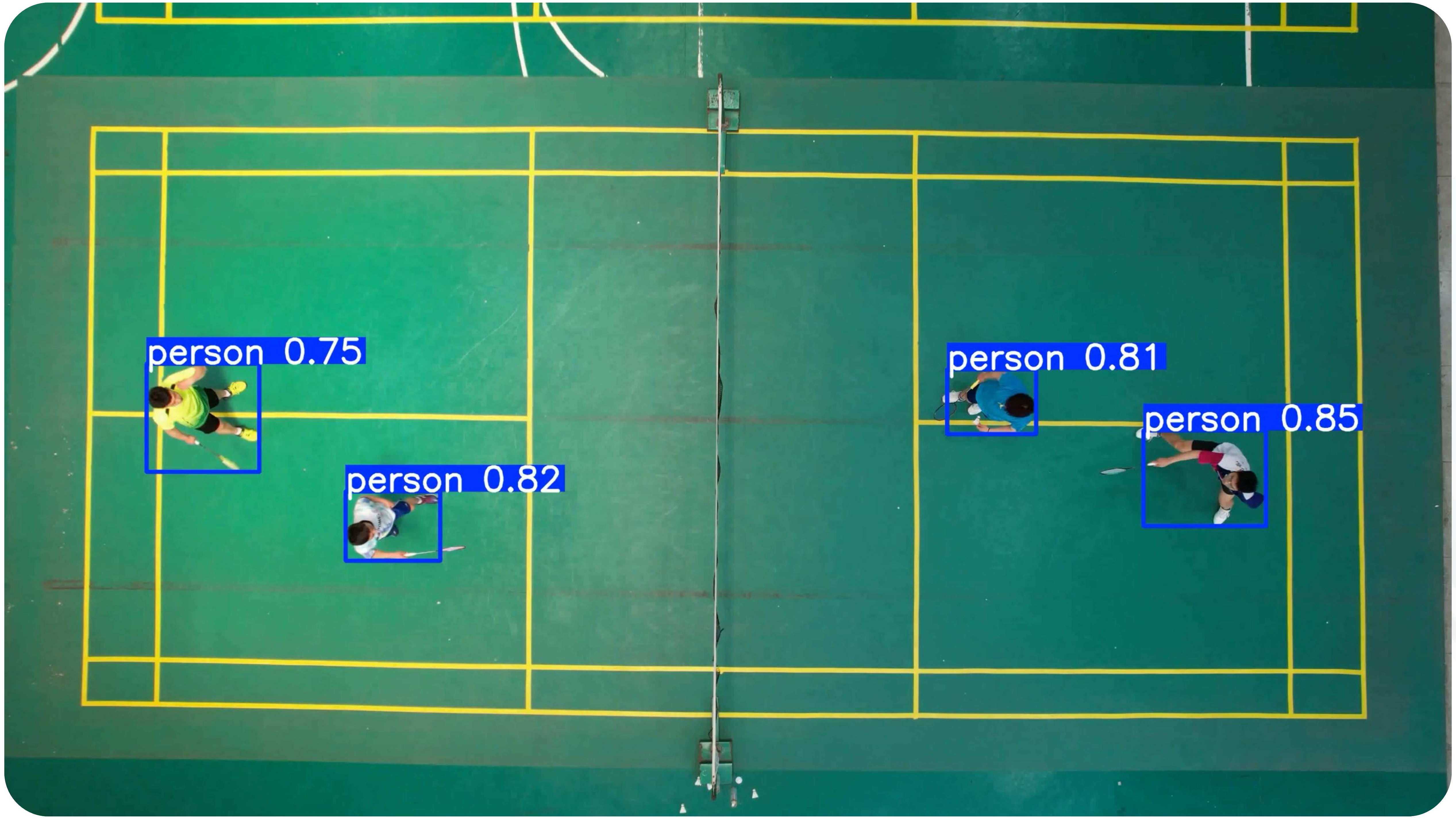
Figure 3.4. Example of Image Mixup that generated by linearly combining pixel values from two input images.

Performance comparison of player detection on the COCO dataset based on mAP@50–95.

Various Heuristics	mAP	Δ	Cumu Δ
Baseline	33.05	0	0
+ Data Pre-processing	33.58	0.53	+ 0.53
+ Cosine LR Schedule	34.52	0.94	+ 1.47
+ SyncBN	35.67	1.15	+ 2.62
+ Image Mixup	37.35	1.68	+ 4.3

Table 1. mAP comparison after various heuristics training.

Baseline + Various Training Heuristic



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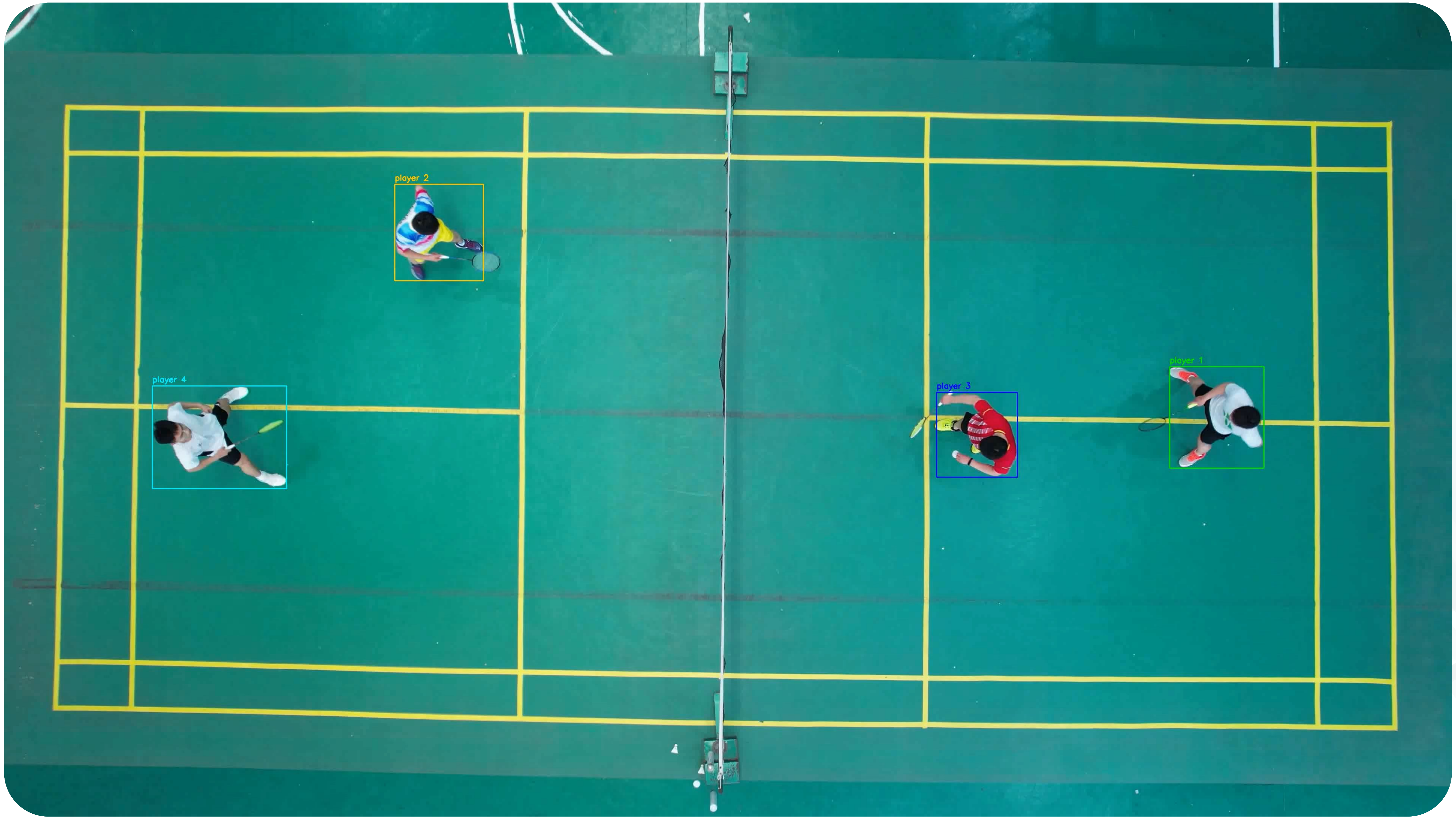
Baseline + Various Training Heuristic + SORT (Tracking)



Player Detection

Result : Player Heatmap Statistics

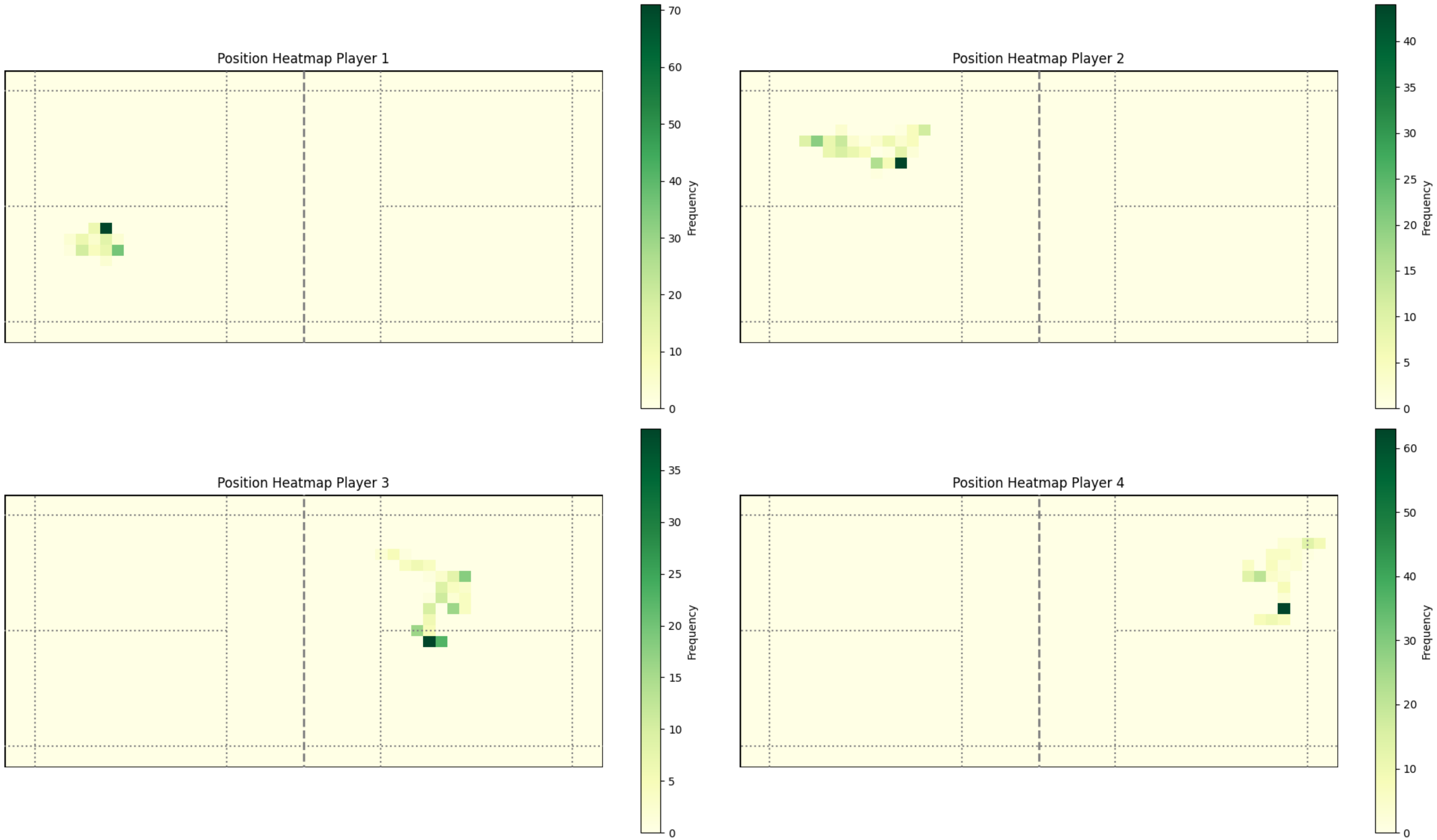


Figure 6.3. position heatmap for each player in badminton double matches.

Player Detection

Result : Player Speed Statistics

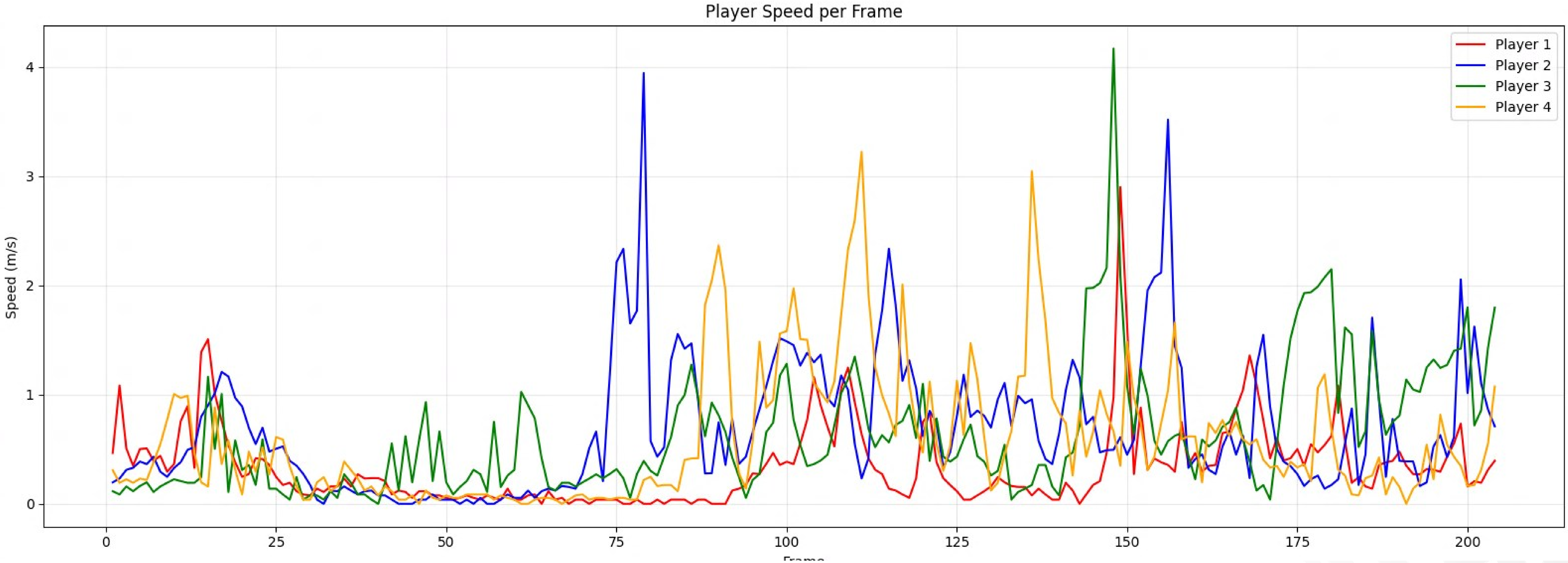
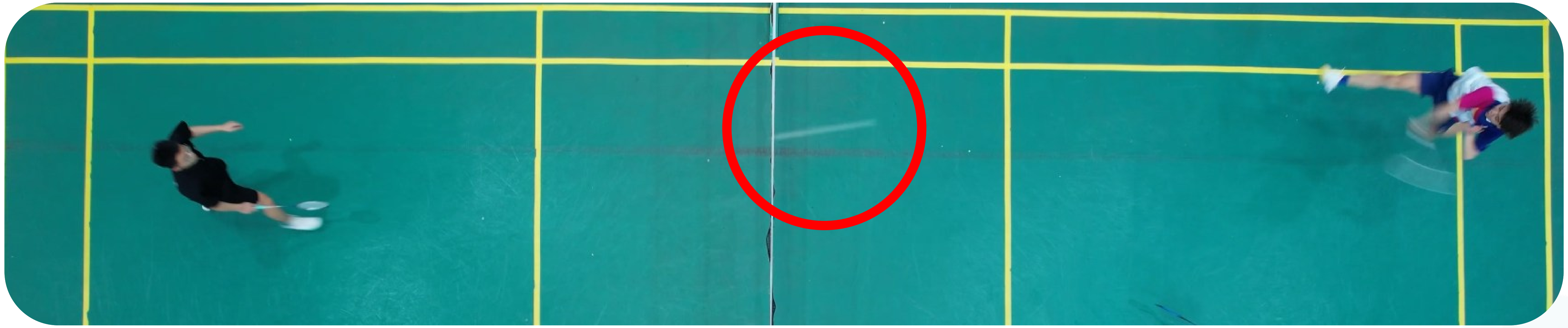


Figure 6.4. Speed statistics for each player in badminton double matches.

Outlines

Shuttlecock Detection

Introduction



- Shuttlecock detection is a challenging task due to its small size, fast motion, and frequent motion blur in badminton videos, which often lead to unstable predictions. Even with a strong baseline (TrackNetV2), detection performance can degrade in fast rallies and complex motion patterns.

- To address this problem, this study builds upon TrackNetV2 as a baseline by introducing residual connections, replacing ReLU with Leaky ReLU, and repositioning batch normalization after residual addition. These modifications aim to tackle vanishing gradient problem and improve training stability.

Shuttlecock Detection

Method : Residual Connection

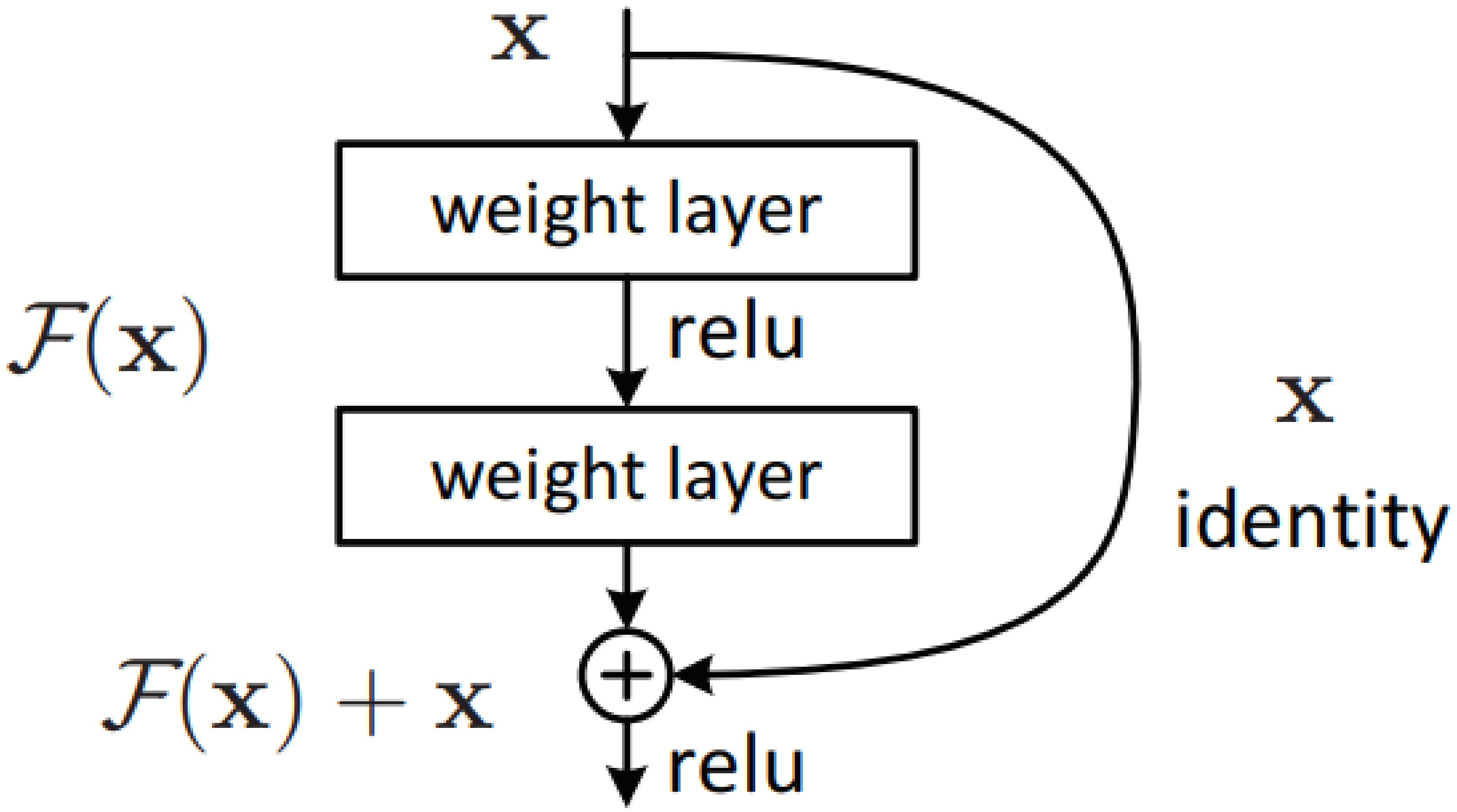


Figure 4.1. Residual block architecture illustrating the skip connection, where the input x is added to the residual mapping $F(x)$

Shuttlecock Detection

Method : Residual Connection (Cont.)

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Proposed the extended version of baseline (TrackNetV2) by introducing residual connections (red arrow) in its blocks. Residual connection could mitigate the gradient vanishing problem, resulting in higher shuttlecock detection performance.

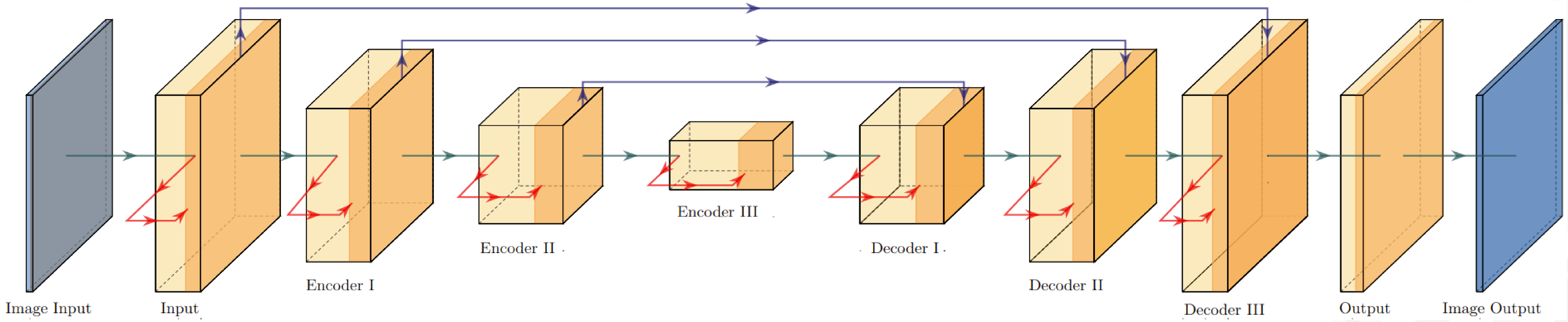
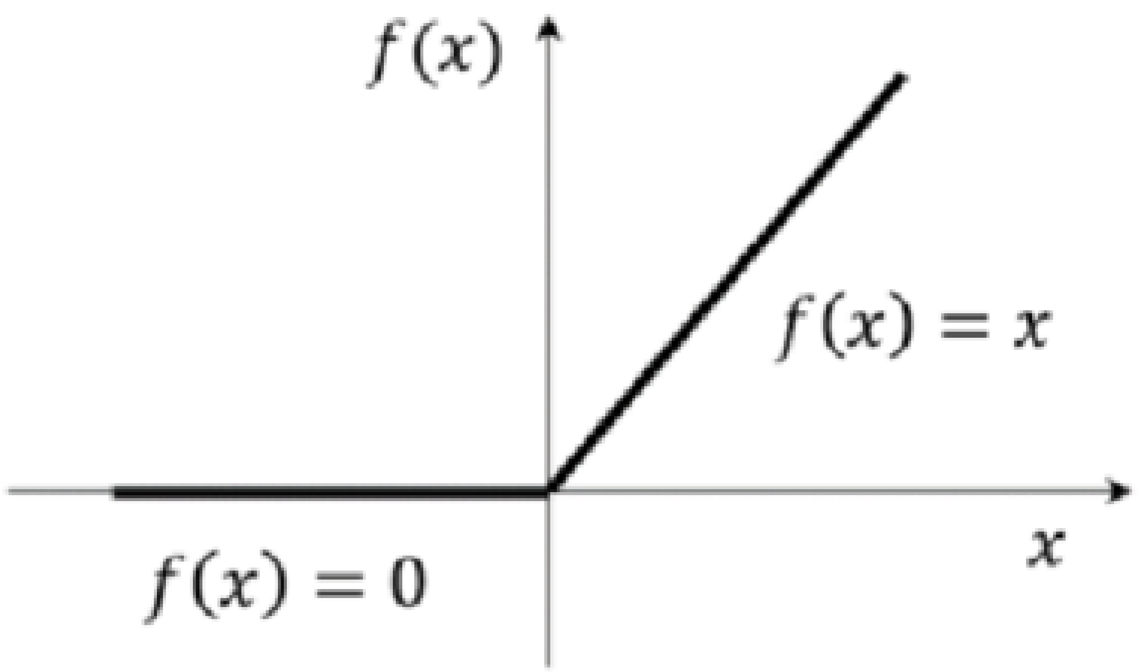


Figure 4.2. Overall network layer detail of shuttlecock detection system with the addition of residual function.

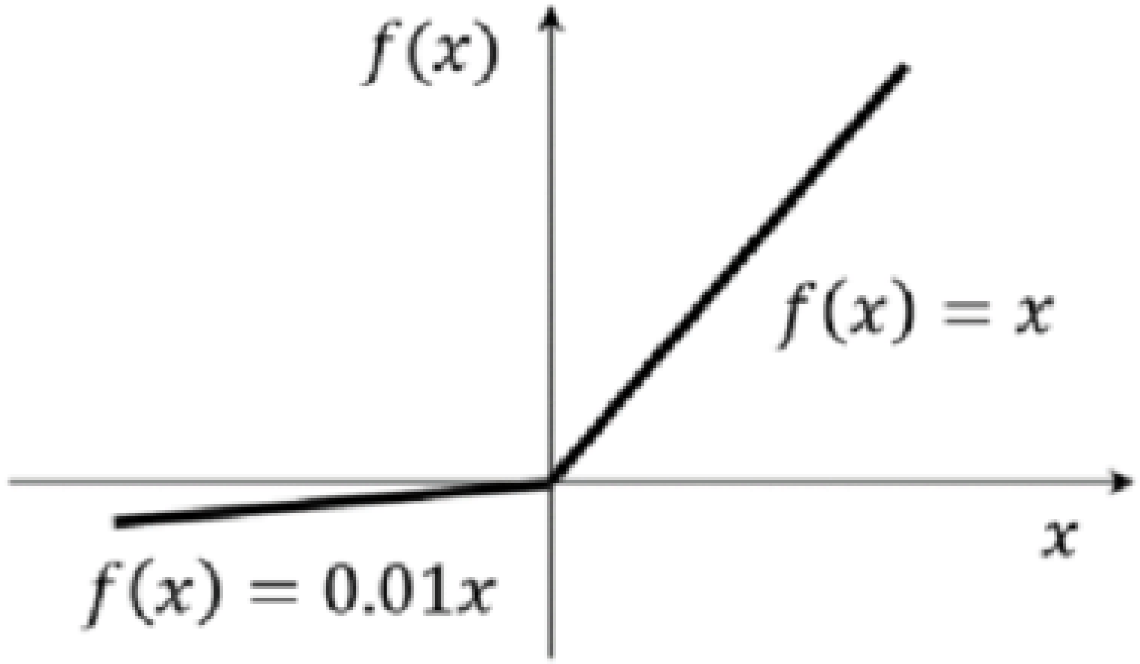
Shuttlecock Detection

Method : LReLU

The shuttlecock is small, fast, and often appears faint or partially blurred. ReLU suppresses negative activations, which can remove weak but important features. This increases missed detections when the shuttlecock signal is subtle.



ReLU activation function



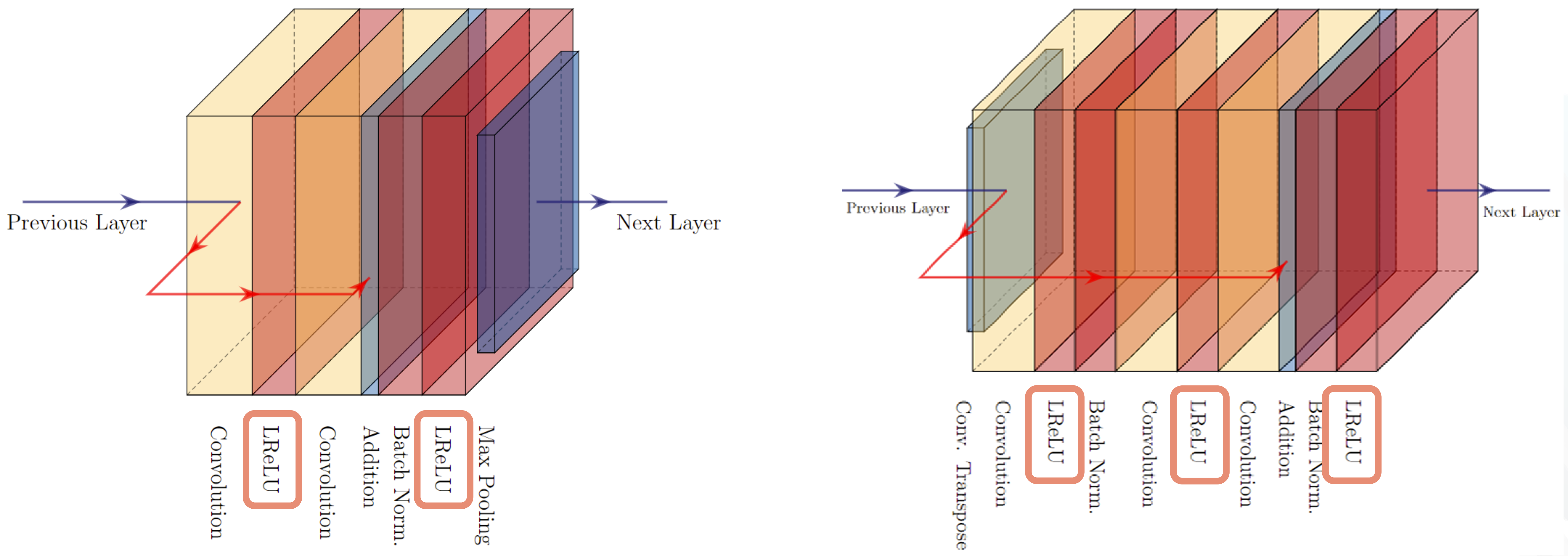
LeakyReLU activation function

Chapter 4

Shuttlecock Detection

Method : LReLU (Cont.)

Leaky Rectified Linear Unit is used to address the dying ReLU problem by preventing some neurons from turning inactive and stopping learning during training.



(a) (b)

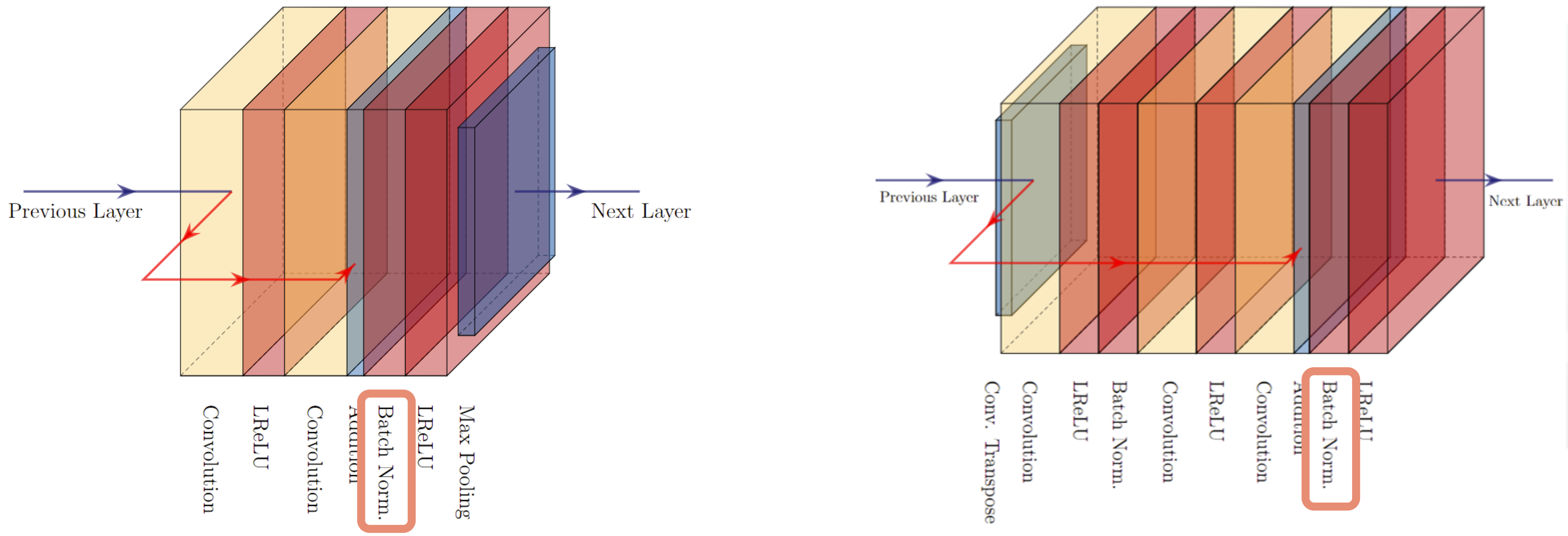
Figure 4.3. Encoder (a) and decoder (b) block in with Leaky ReLU implemented.

Chapter 4

Shuttlecock Detection

Method : Post-residual Batch Normalization

Batch normalization (BN) layers are early in the network to normalize the activations. However, it may cause over-normalization, which could degrade the shuttlecock detection performance. By applying BN after the addition operation, the input to the activation function is normalized.



(a) (b)

Figure 4.4. Encoder (a) and decoder (b) block with batch normalization placed after residual block.

Chapter 4

Shuttlecock Detection

Dataset

Figure 4.5. Example frames in the shuttlecock detection dataset.

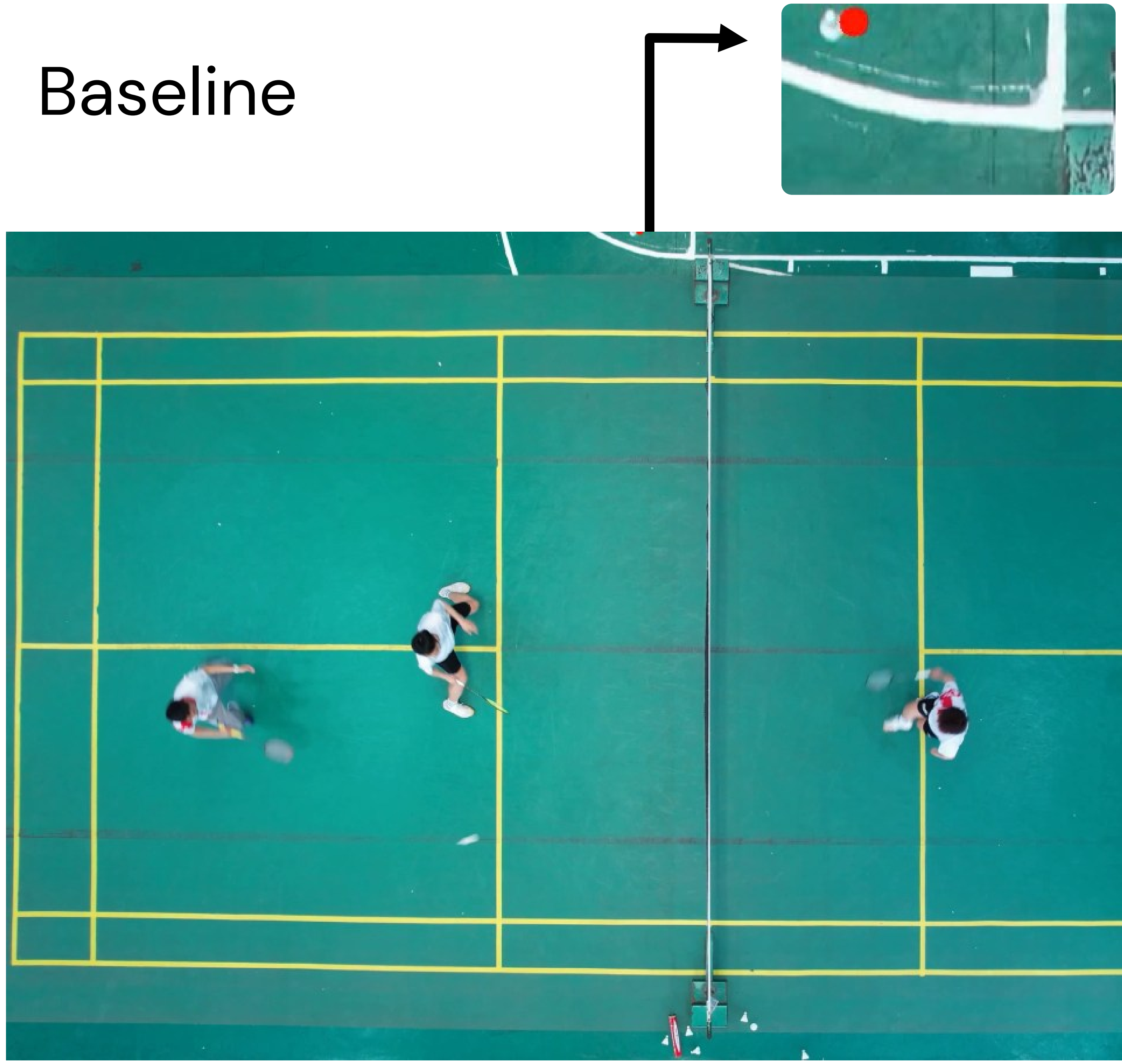
Figure 4.6. Example frames in the shuttlecock detection for testing data.

Chapter 4

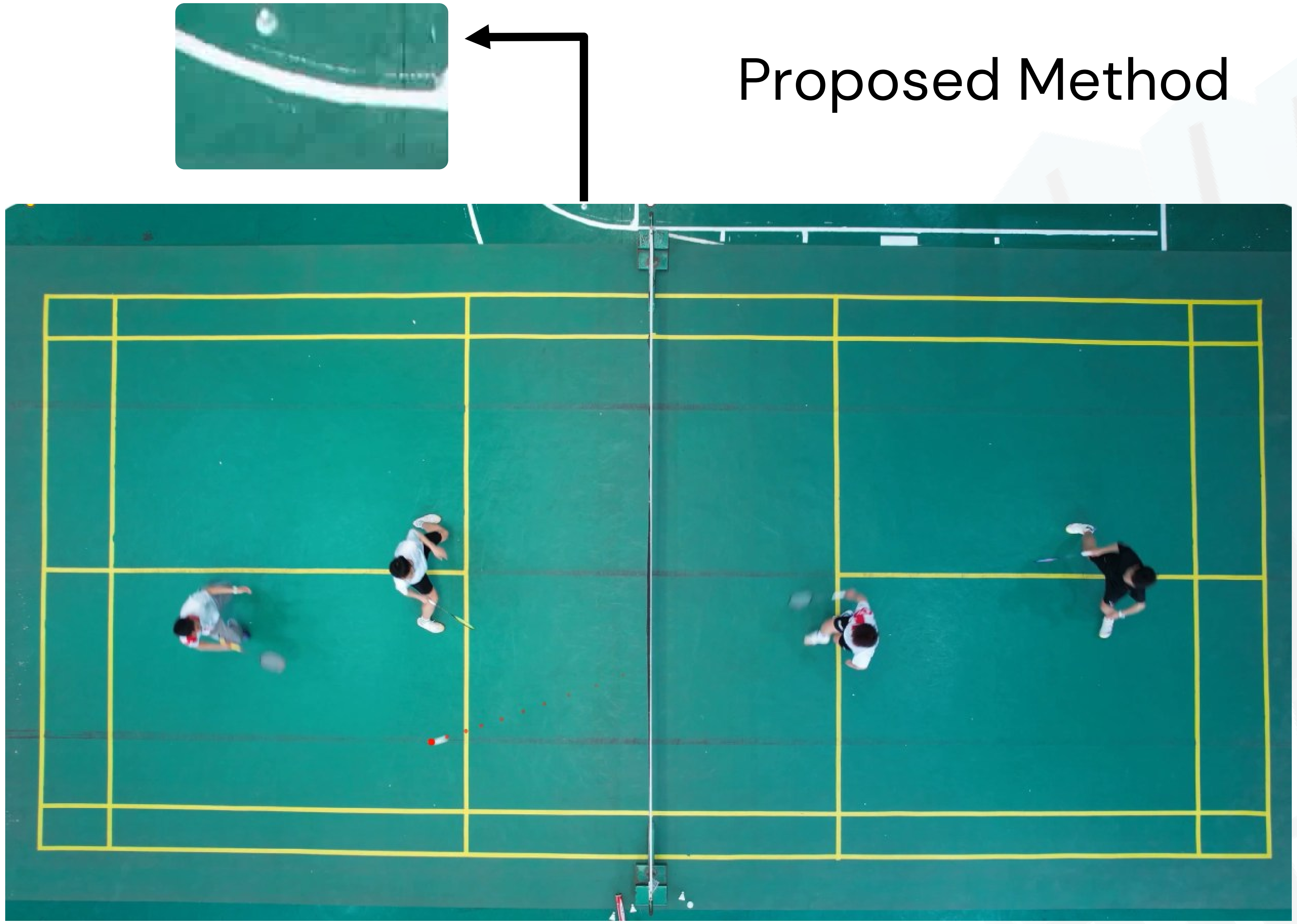
Shuttlecock Detection

Result

Baseline



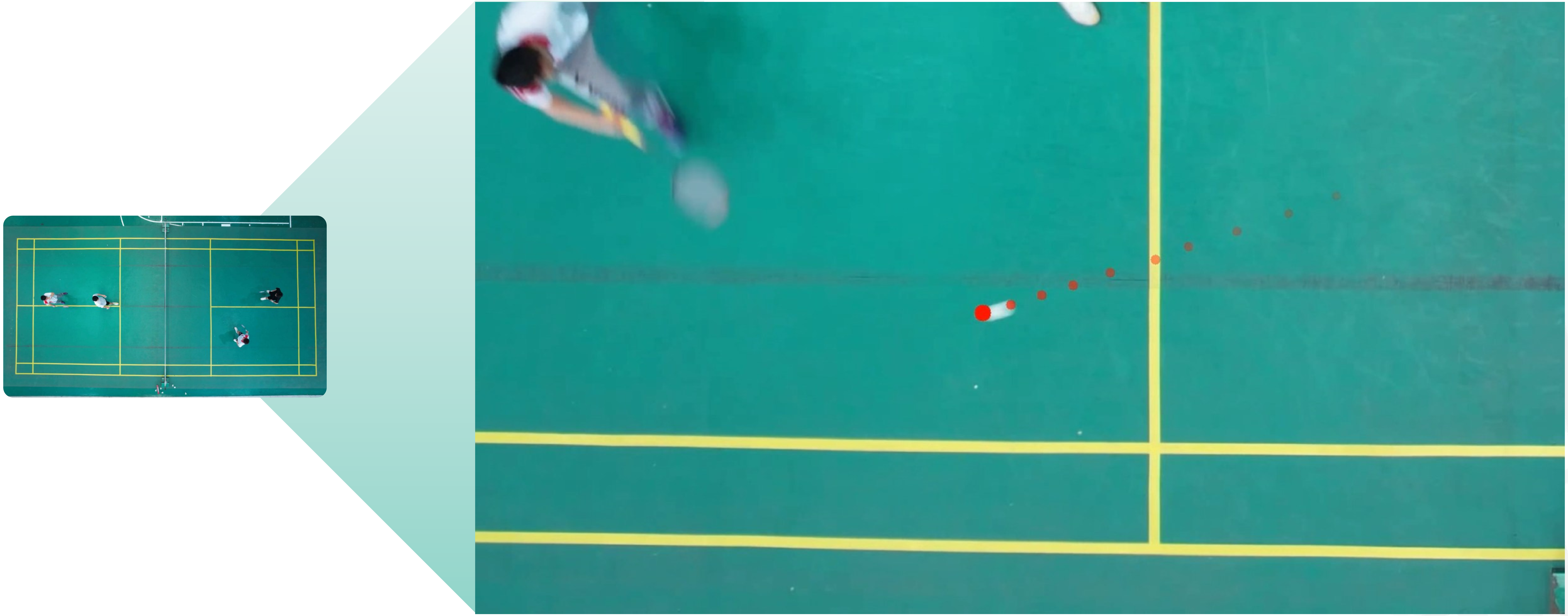
Proposed Method



Chapter 4

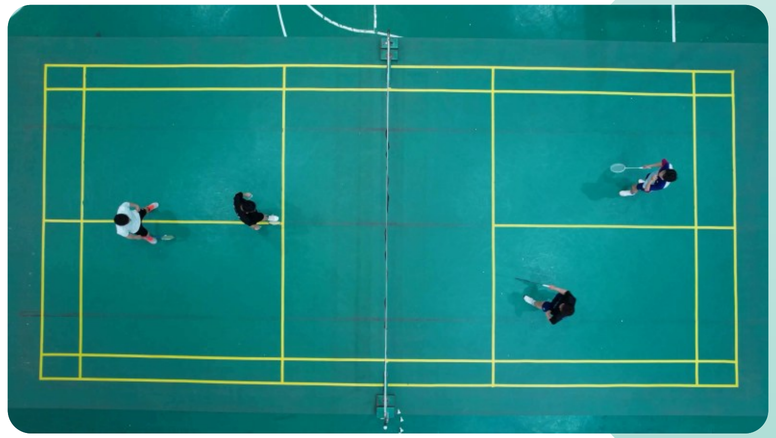
Shuttlecock Detection

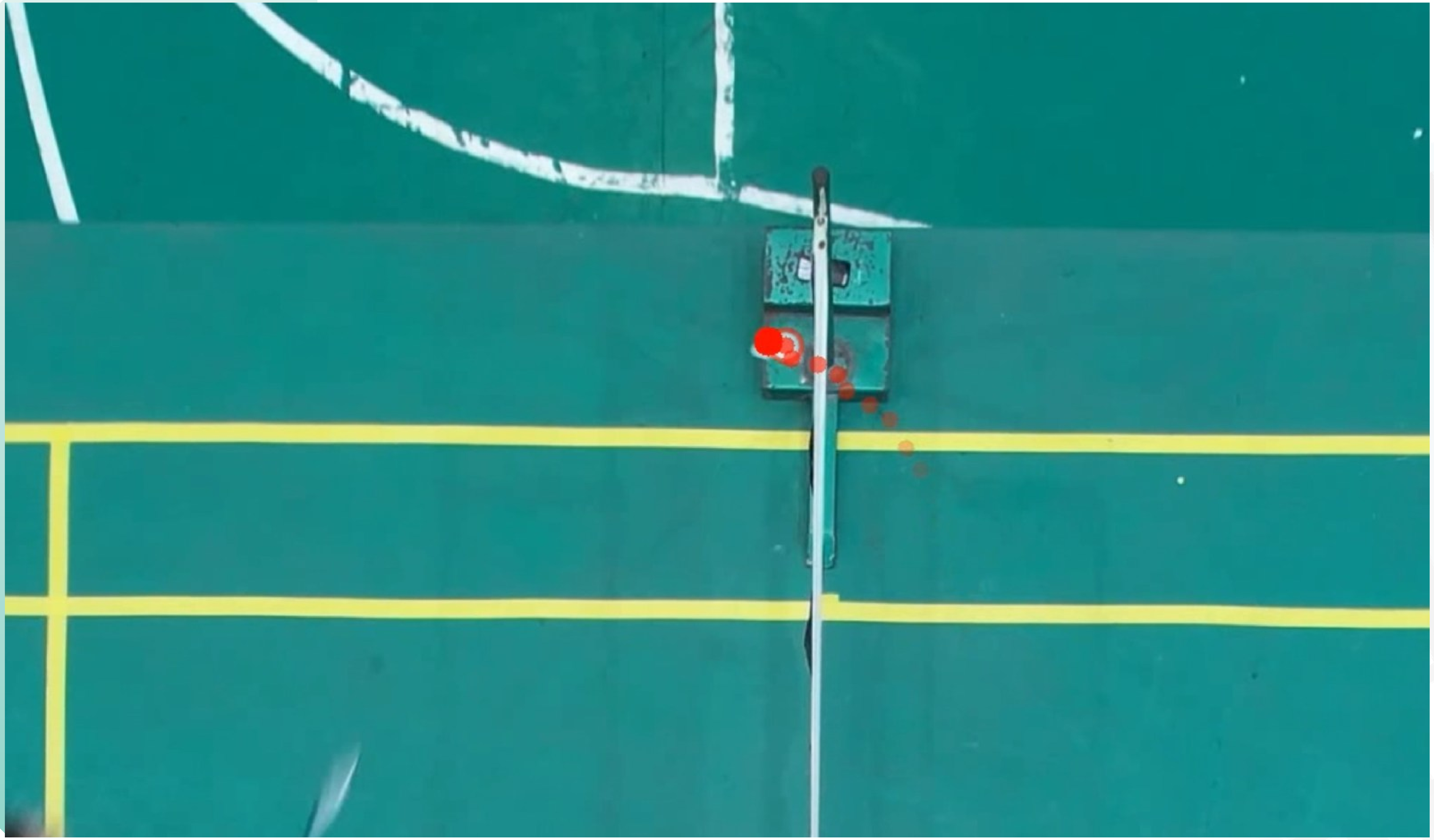
Result (Cont.)



Shuttlecock Detection

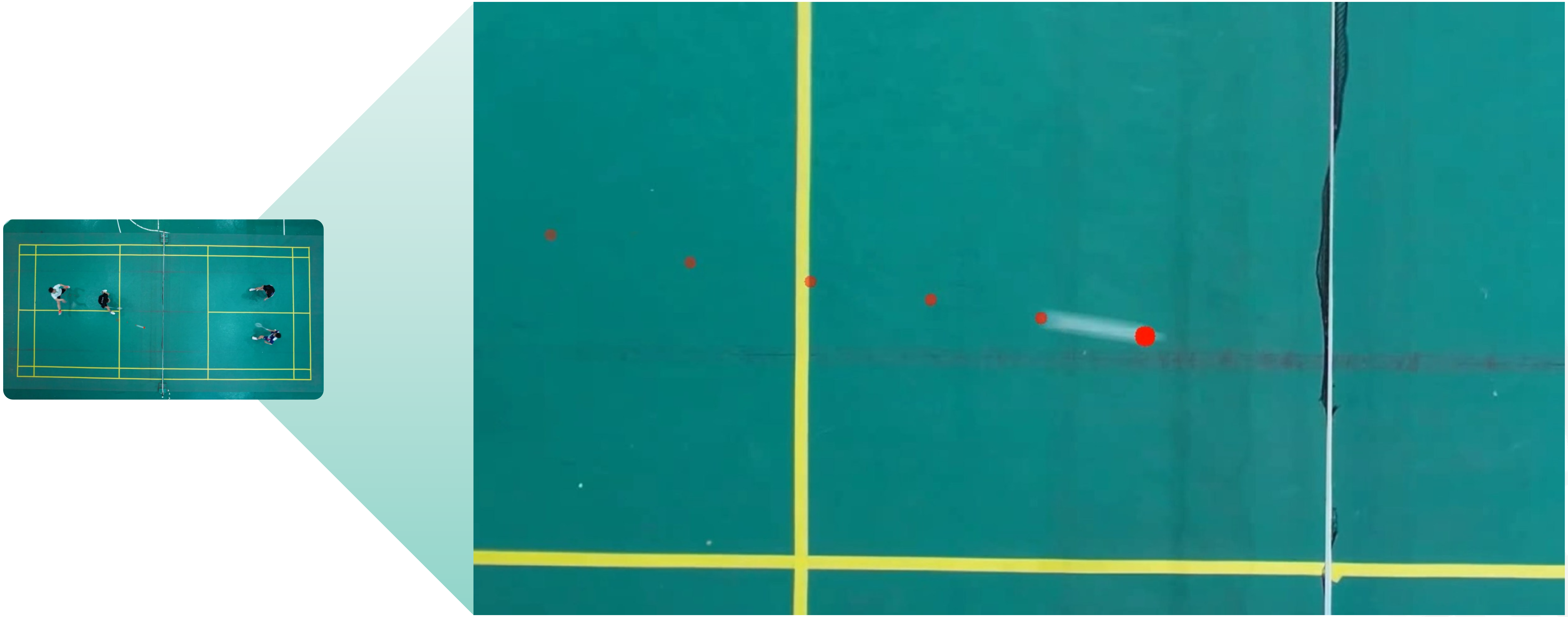
Result : Slow moving shuttlecock (Net Play)





Shuttlecock Detection

Result : Fast moving shuttlecock (Smash)



Shuttlecock Detection

Result : Training Loss

Figure 4.6. Training loss comparison between baseline and proposed method.

Shuttlecock Detection

Result : Metric Comparison

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		Predicted: Negative	Predicted: Positive
Actual	Negative	True Negative (TN)	False Positive (FP)
	Positive	False Negative (FN)	True Positive (TP)

$$Precision = \frac{TP}{TP + FP}$$

$$Recall = \frac{TP}{TP + FN}$$

$$Accuracy = \frac{TP + TN}{TP + FP + TN + FN}$$

$$F1 = 2\,x\,\frac{P\;x\;R}{P + R}$$

Model	Prec.	Rec.	F1	Acc.
Baseline	84.8%	81.8%	83.3%	75.8%
+ Residual network	87.4%	82.0%	84.6%	77.6%
+ Leaky ReLU	87.9%	82.6%	85.0%	77.9%
+ Post-Addition BN	88.2%	83.1%	85.3%	78.1%

Table 4.1. Performance comparison between baseline with proposed method.

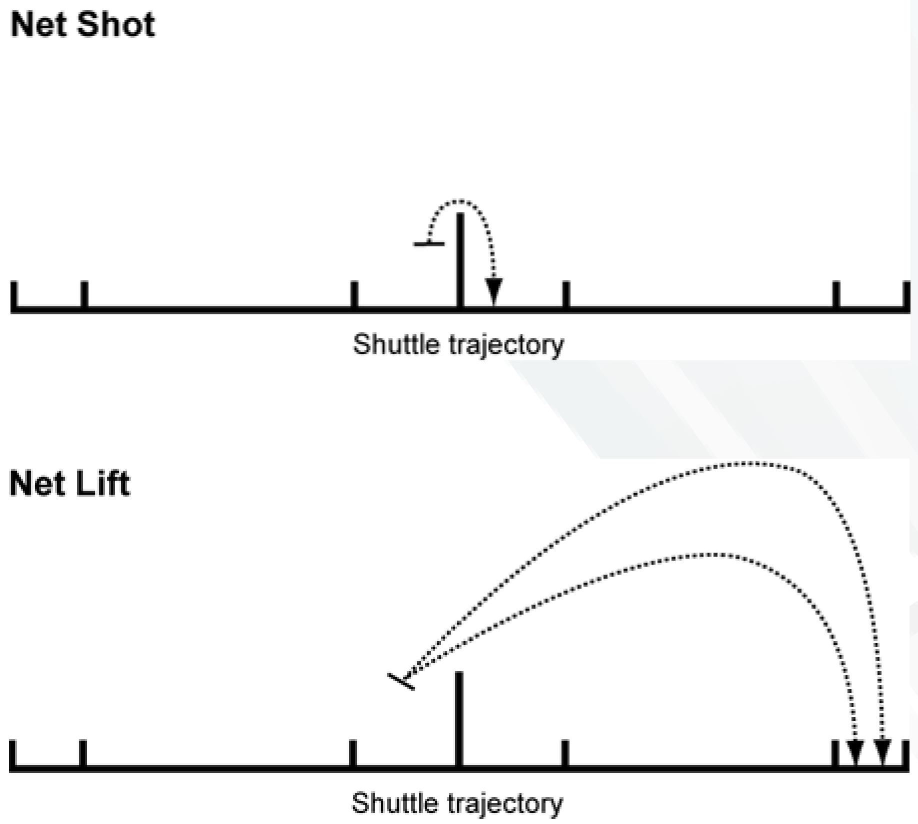
Outlines

Racket Detection

Introduction

Why racket matter?

The racket defines the effective interaction point between player and shuttlecock. In net play, small changes in racket position strongly affect shot outcome. Racket orientation reflects intention, timing, and shot readiness.



Racket Detection

Introduction (Cont.)

- Racket detection in badminton videos is challenging due to the lack of publicly available datasets that provide explicit racket orientation information. Existing badminton datasets mainly focus on players or shuttlecocks, making racket direction analysis difficult to perform reliably.

- To address this problem, this study constructs a dedicated badminton racket dataset with oriented bounding box annotations, including racket angles.

- Racket detection is conducted using YOLOv11-OBB, and SORT is applied to maintain temporal consistency across frames.

Racket Detection

Dataset : Source

ORDB (Oriented Racket Database) is composed of a total of 8,704 images taken from double badminton video matches, carefully curated and annotated to provide a diverse range of badminton gameplay scenarios. The dataset is annotated with rotated bounding boxes:

Racket Detection

Dataset : Annotation

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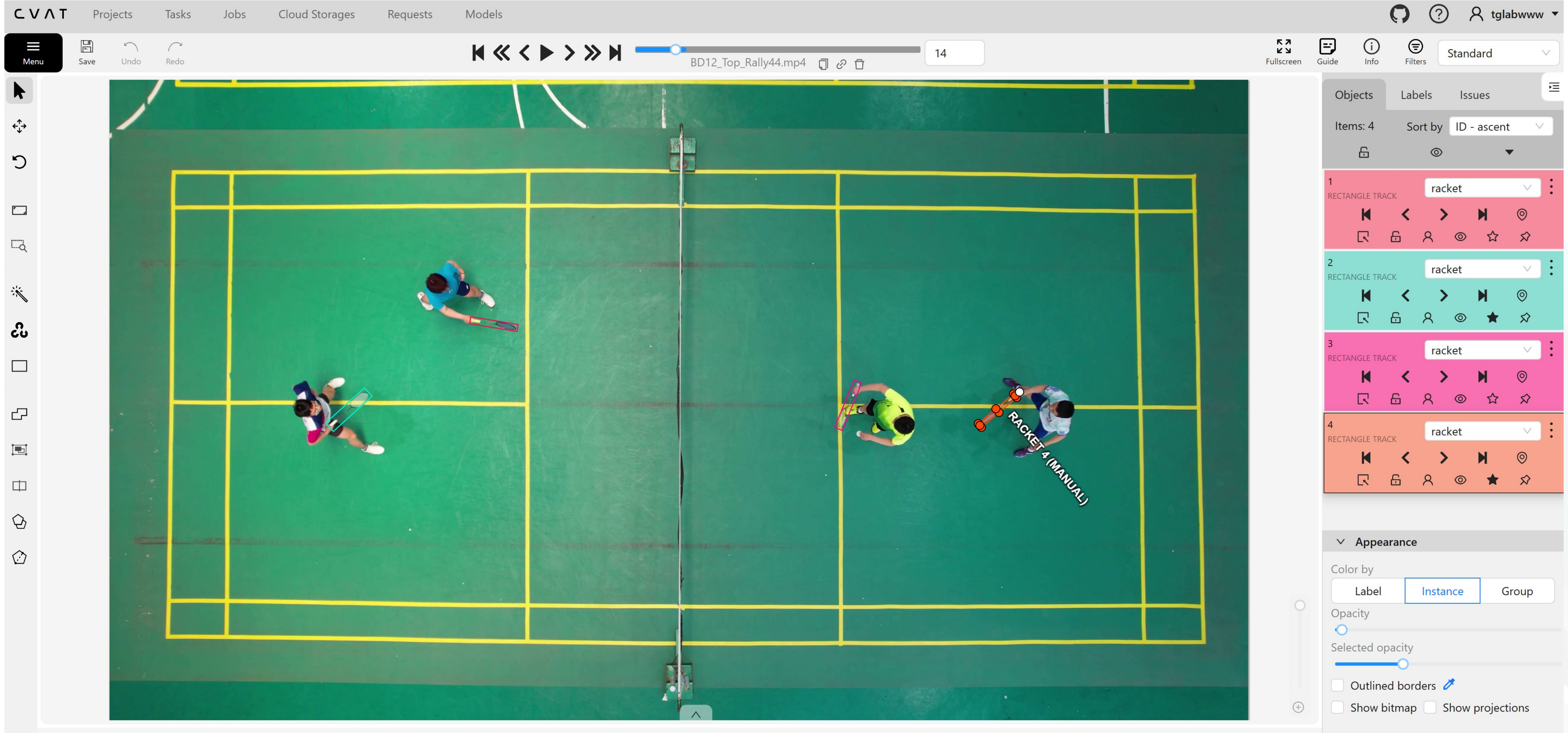

Figure 5.1. Annotation process of oriented racket dataset using CVAT

Racket Detection

Dataset : Data Distribution

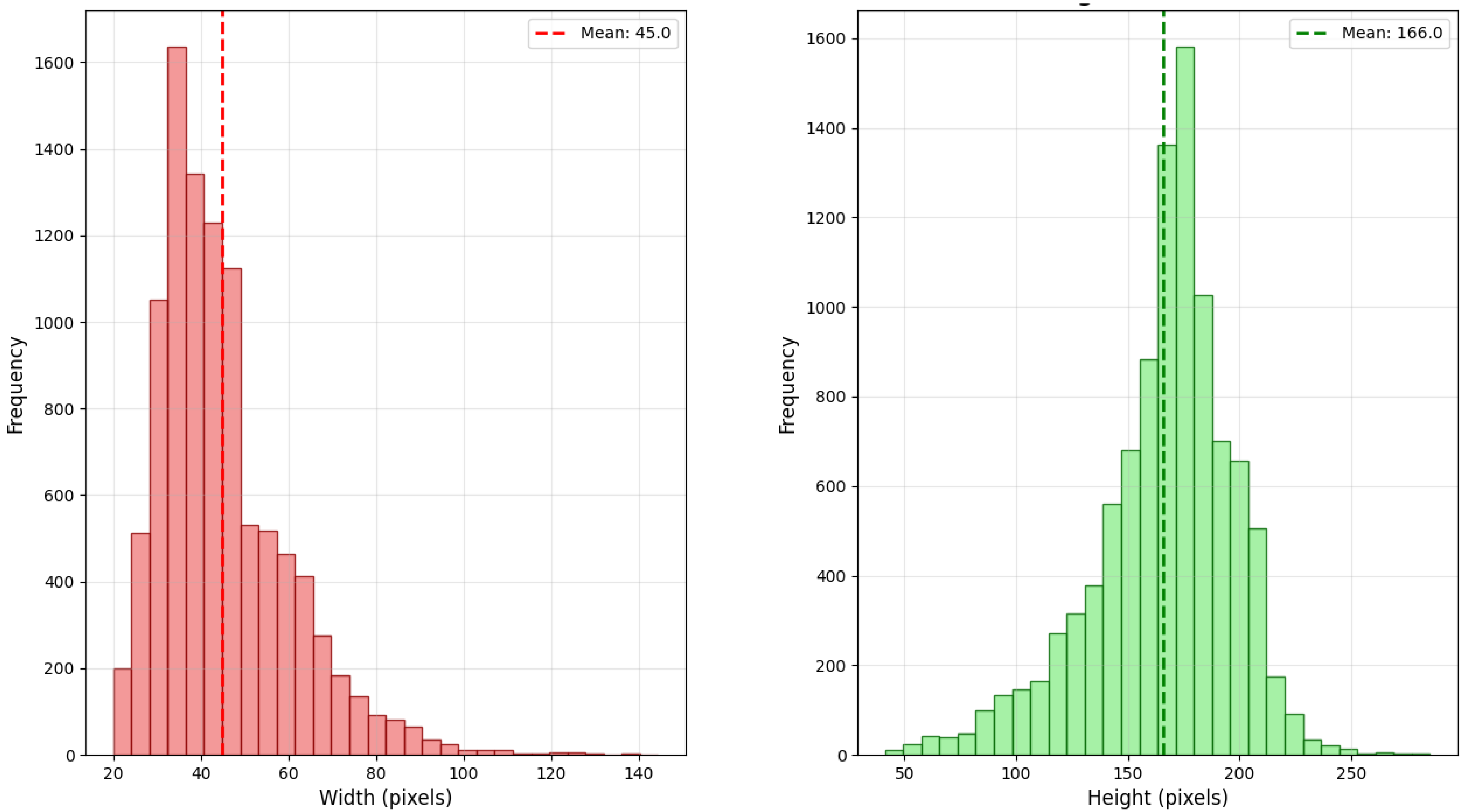

Figure 5.2. Histogram of bounding box width and height

Racket Detection

Dataset : Data Distribution (Cont.)

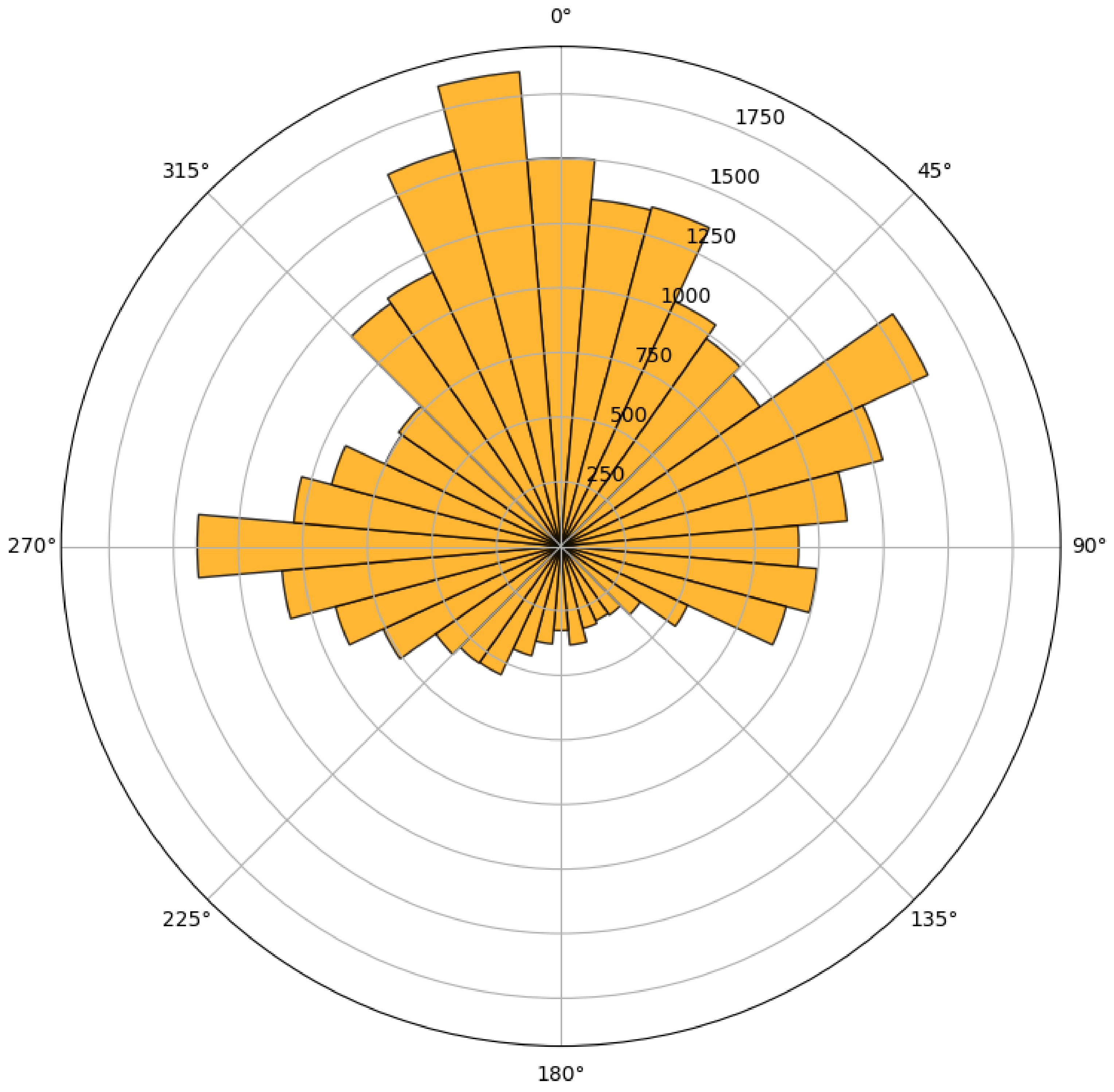

Figure 5.3. Distribution of racket orientation angles in the ORDB dataset.

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YOLOv11 is used for training and inference, specially for OBB (Oriented Bounding Box) version.

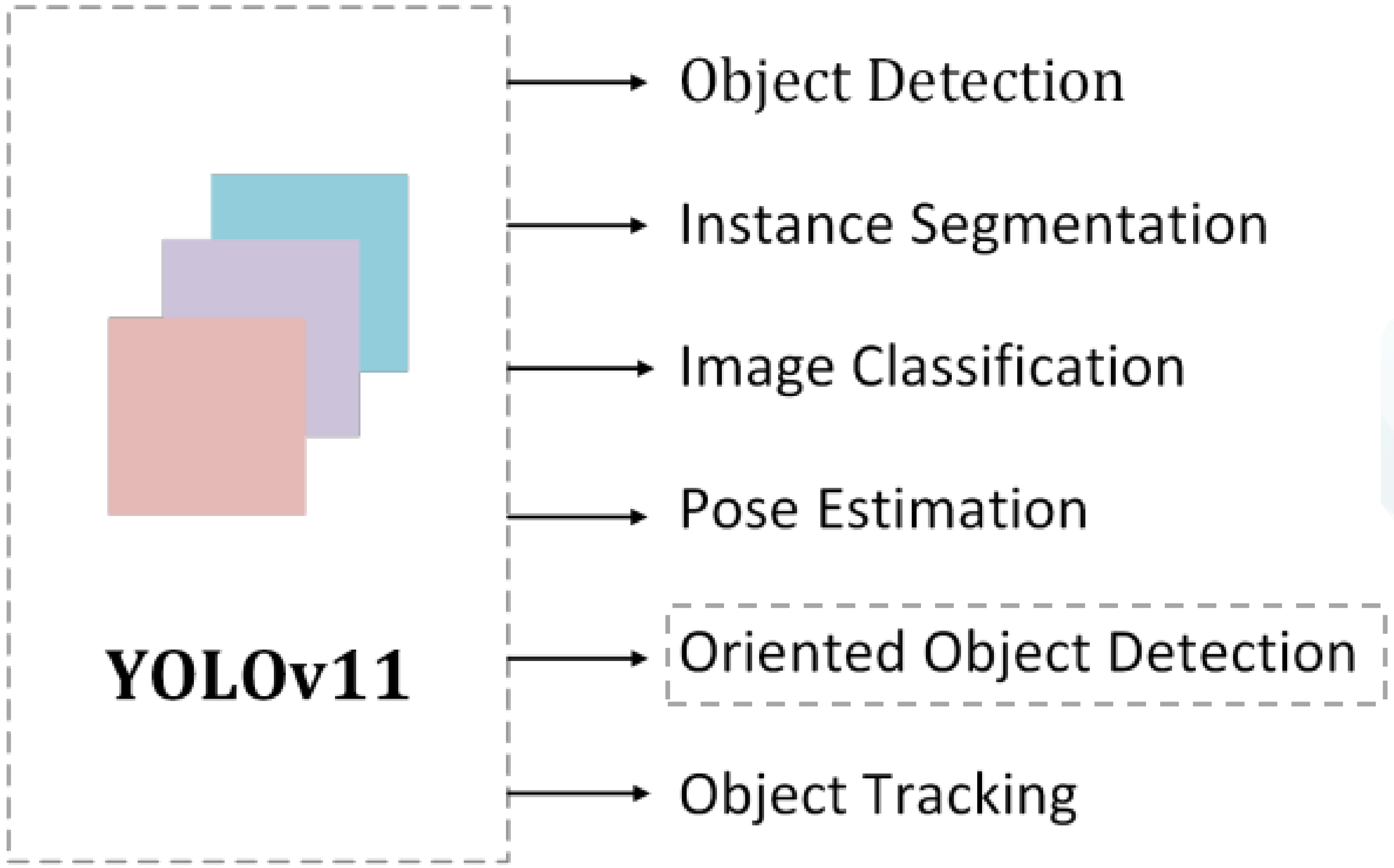

Figure 5.4. YOLOv11, the latest YOLO version as of January 2026, capable of performing oriented object detection.

Racket Detection

Result : YOLOv11 OBB

Qualitative results of YOLOv11-OBB on validation frames. The model successfully detects all four rackets and accurately predicts their positions and orientations.

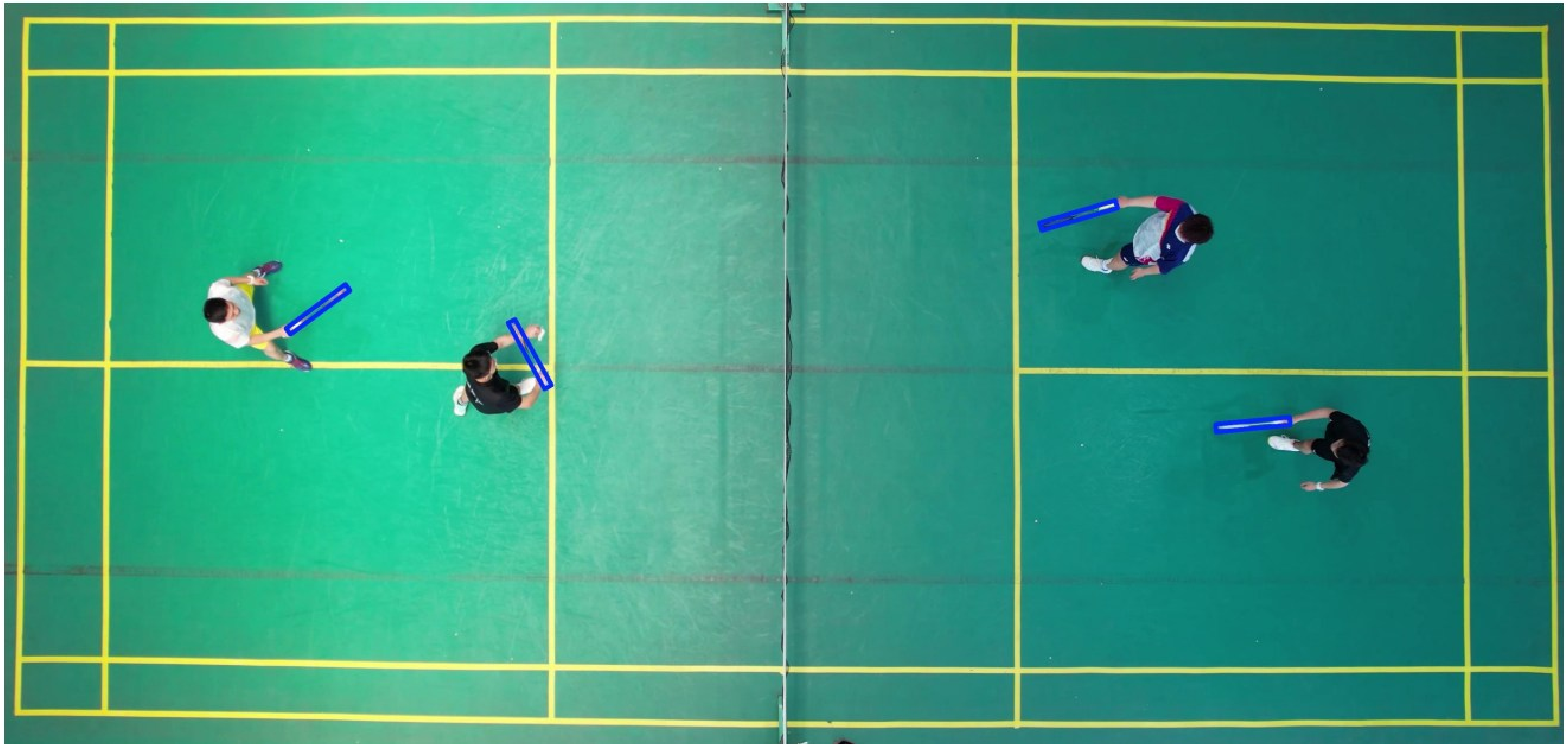

Figure 5.5. The racket was detected successfully with the angle information for every player.

Chapter 5

Racket Detection

Result : YOLOv11 OBB + SORT (Tracking)

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Dominant Area

Introduction

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- In badminton doubles, performance depends on how effectively pairs control and share court space rather than on individual skills alone.
- Rapid role switching during rallies requires coordinated positioning to cover open areas and respond to the shuttlecock. Poor spatial coordination often creates gaps that can be exploited by opponents.

Dominant Area

Introduction (Cont.)

- This study addresses this problem by introducing a dominant area visualization that estimates court regions reachable by each player based on movement dynamics and shuttlecock context, while also incorporating racket angle information.

- Dominant area represents which player can reach a region of the court faster. It captures spatial control that is not visible from positions alone. This information is fundamental for understanding tactics in fast rallies.

Dominant Area

Method

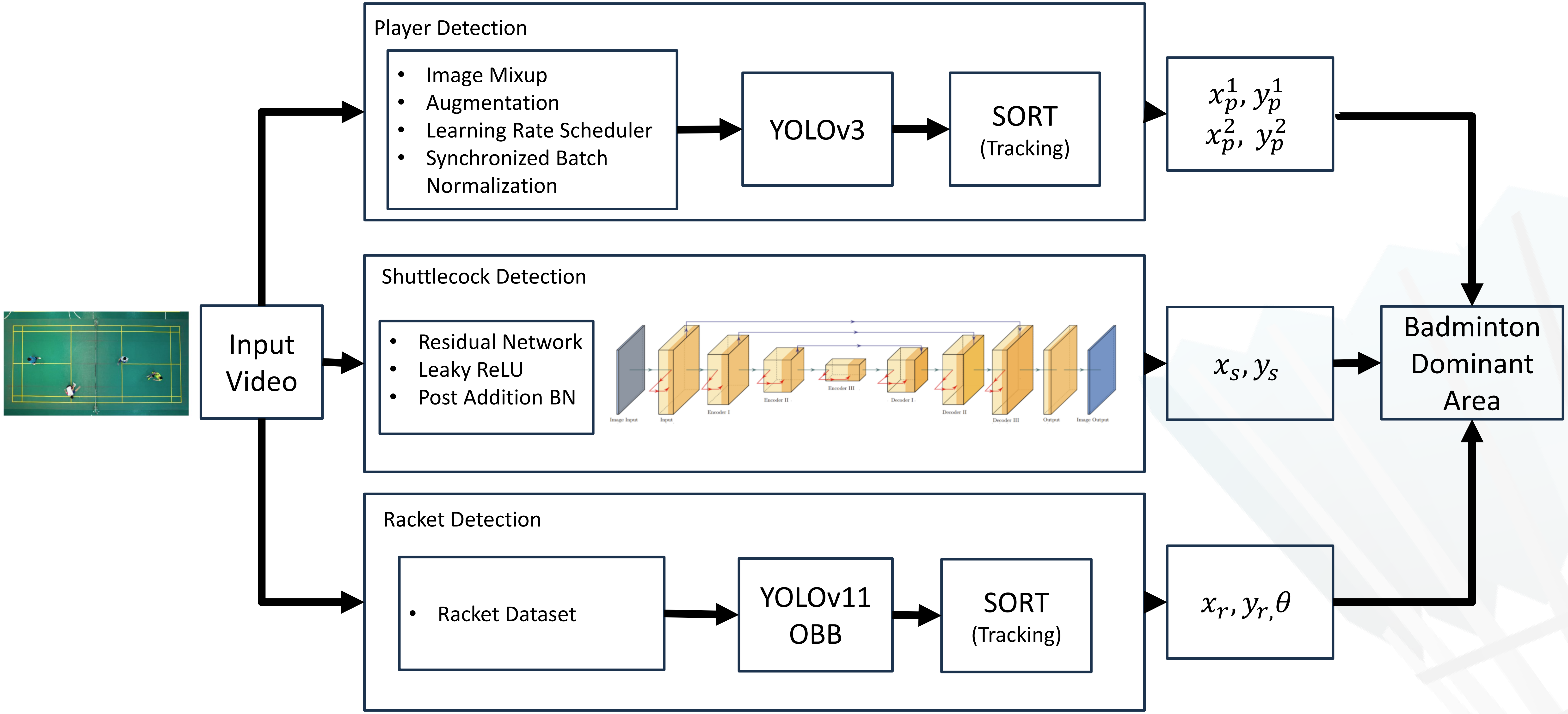
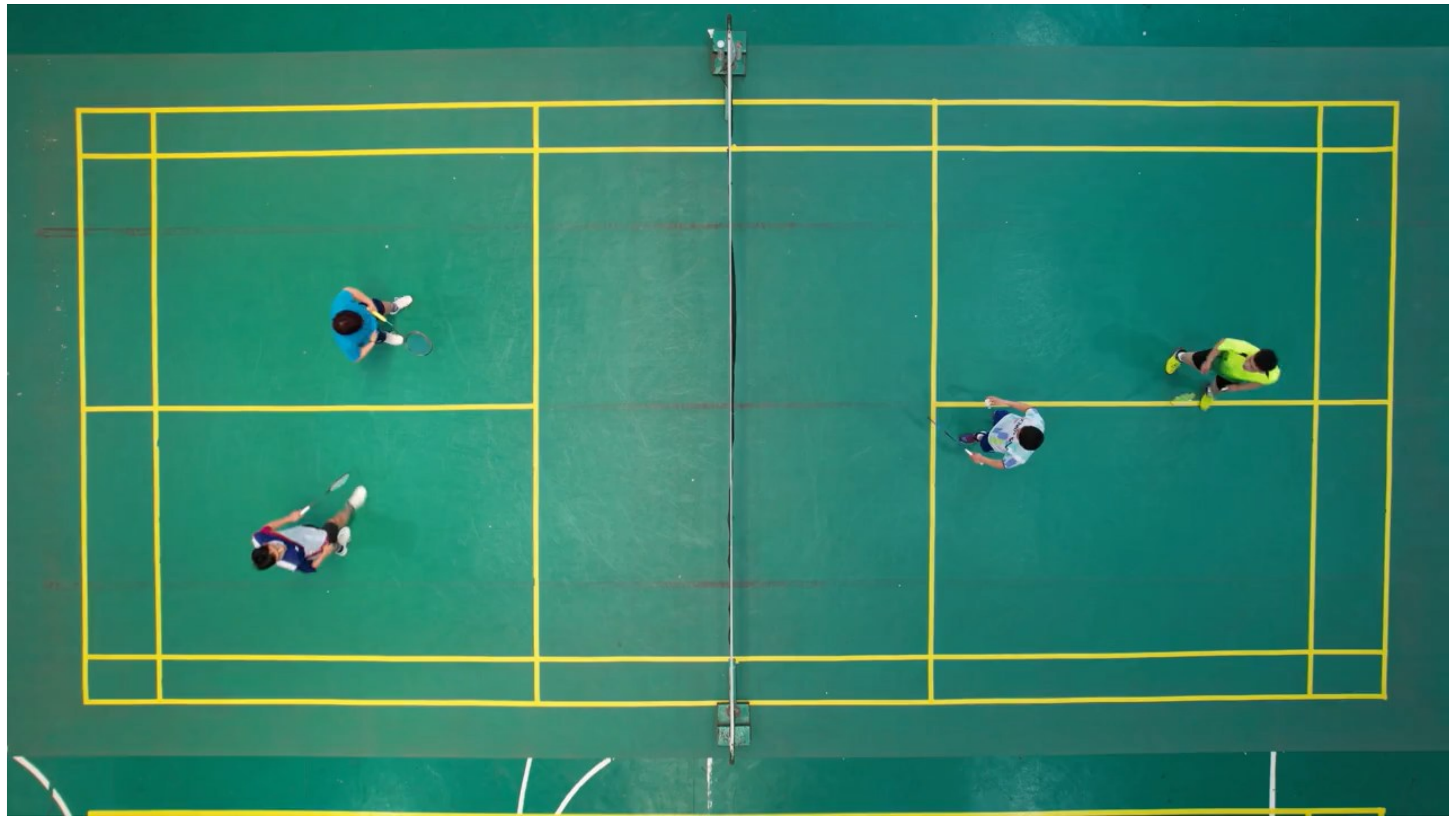
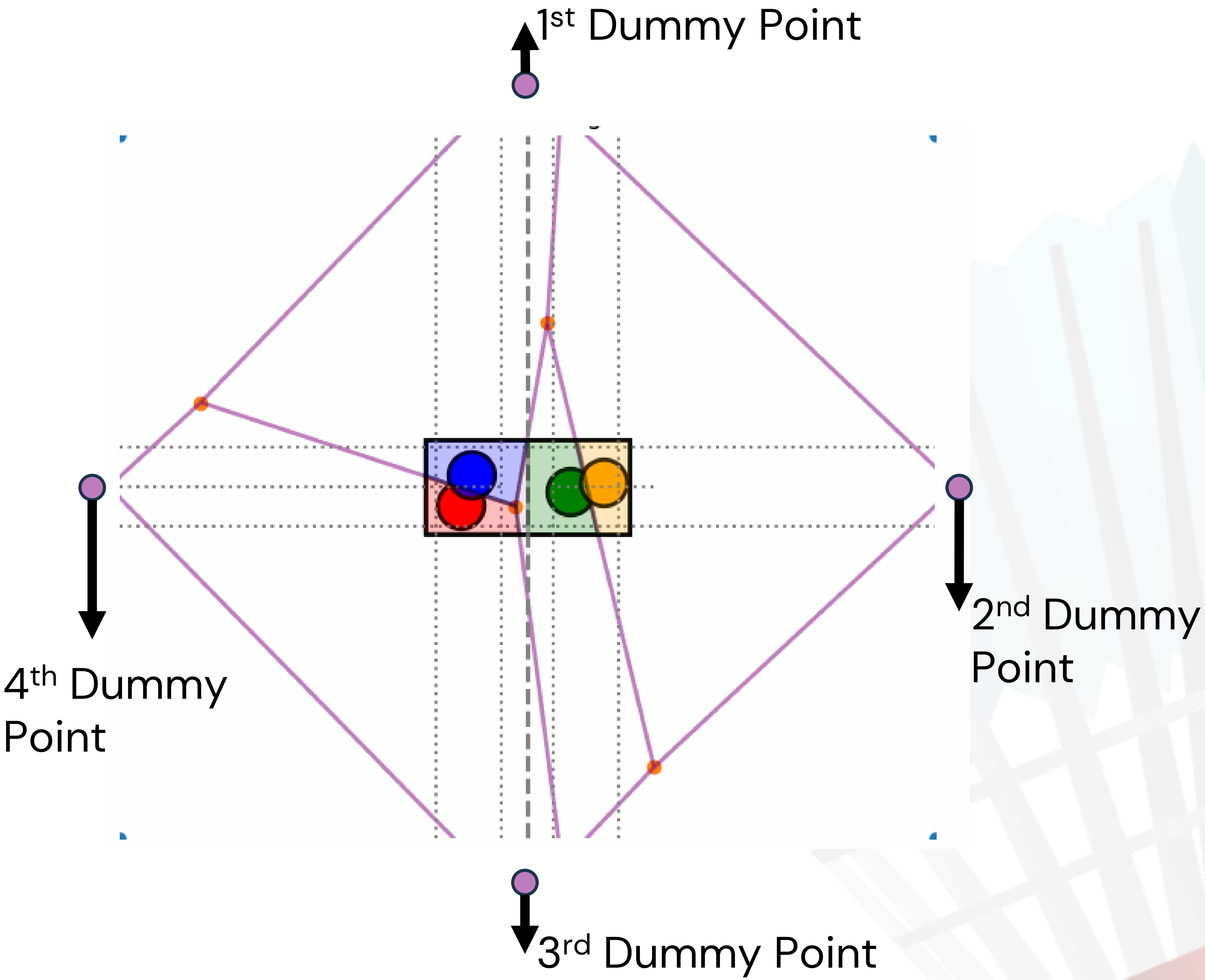


Figure 6.1. Overall pipeline for dominant area estimation in badminton sport analysis

Dominant Area

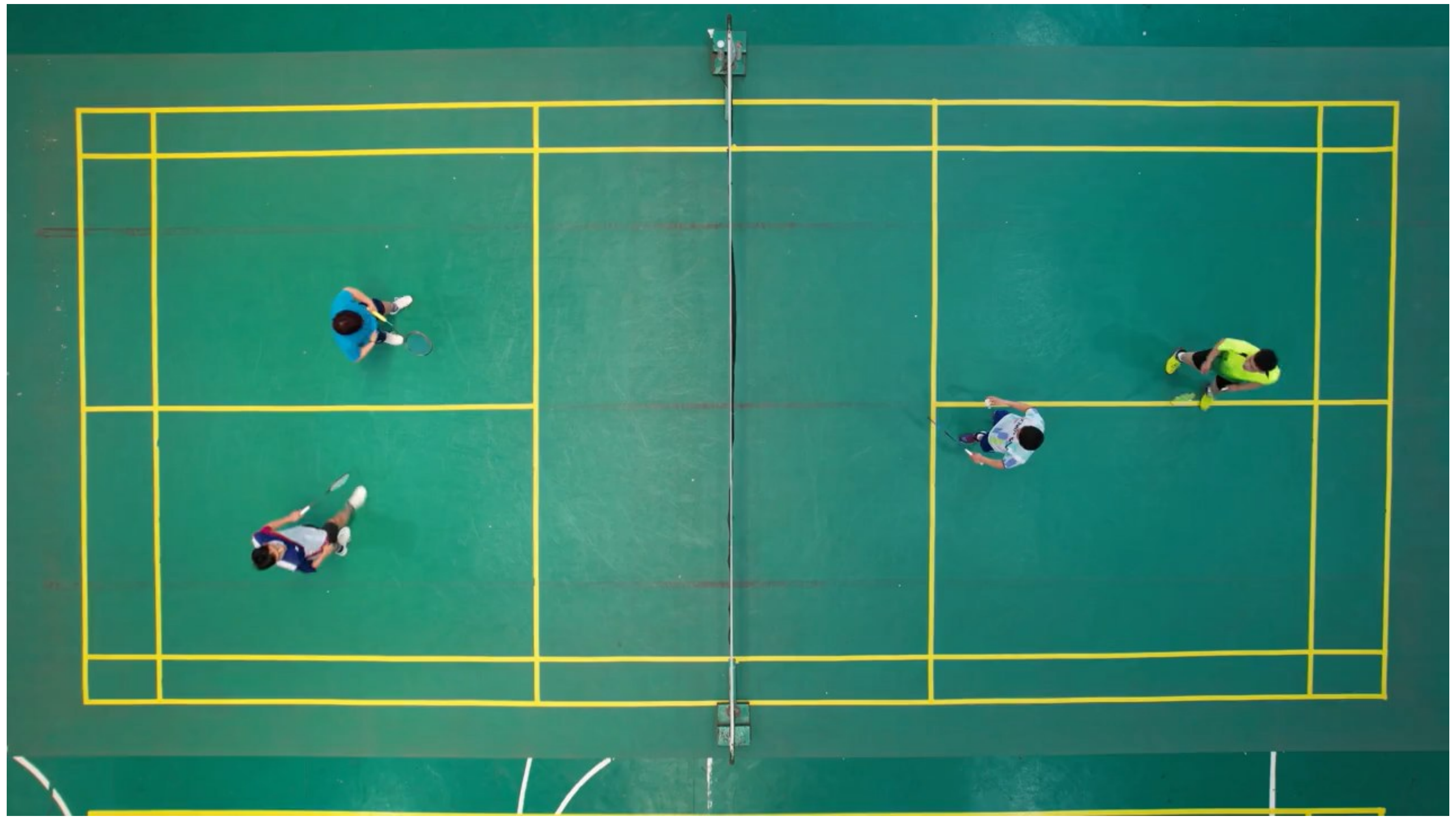
Method : Voronoi Diagram

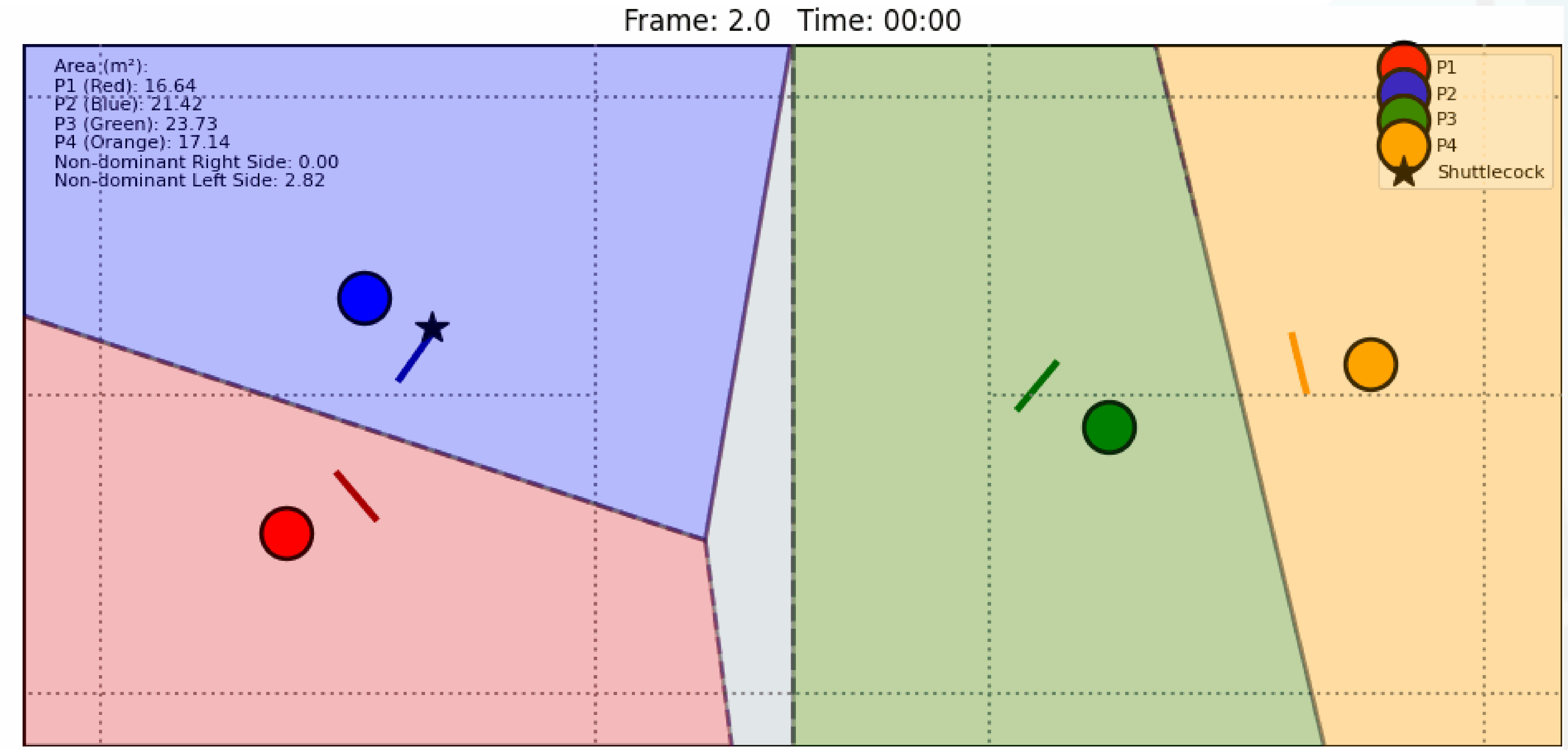




Dominant Area

Result





- In doubles, left–right formation supports defensive coverage.Front–back formation is used for attacking and pressure control.These roles emerge naturally to reduce reaction time and overlap.

Dominant Area

Result : Dominant Area Statistics

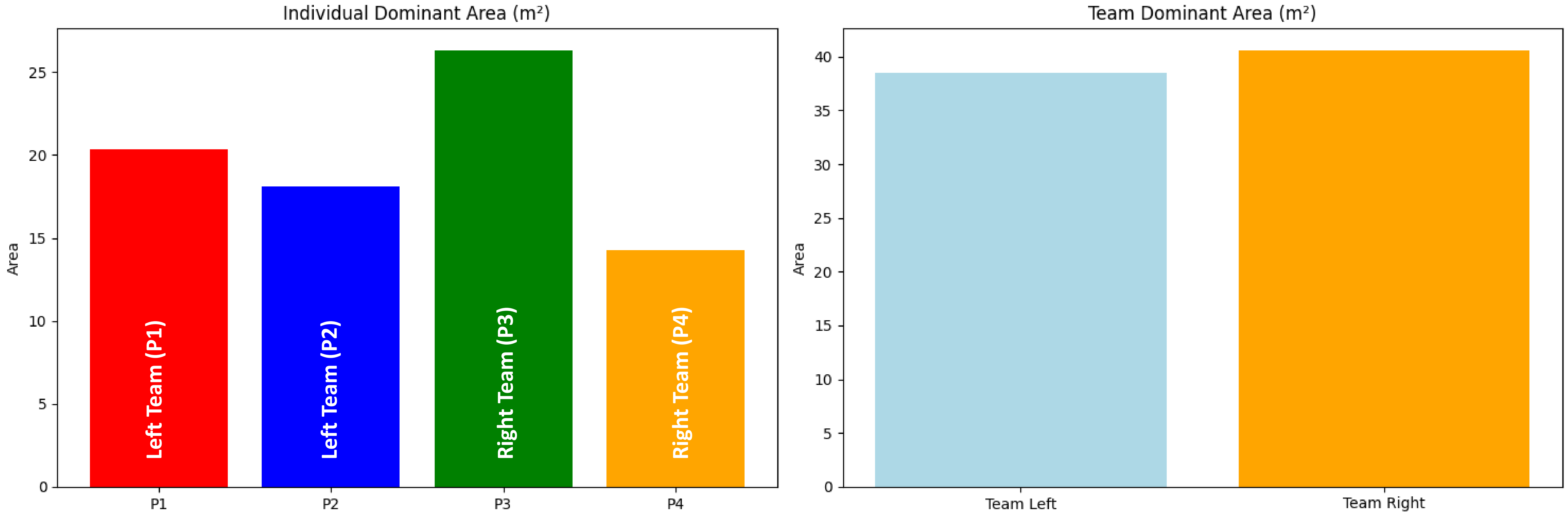


Figure 6.2. Individual and team dominant area in badminton double matches

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Chapter 7

Conclusions and Future Works

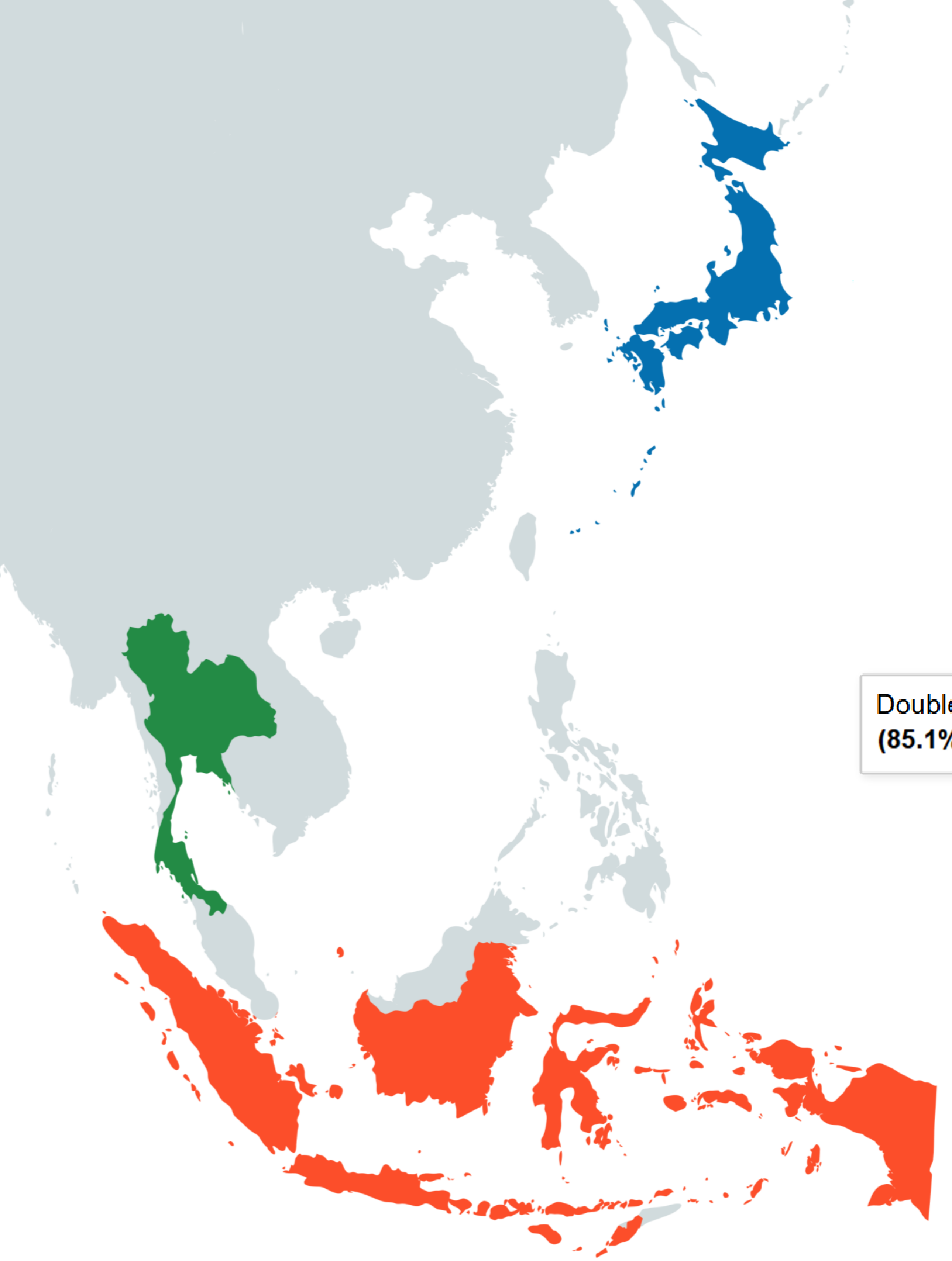
Survey : Questionnaire

- A short user survey was conducted to evaluate the clarity and usefulness of badminton video visualizations.
- Participants rated short rally videos based on player positions, shuttlecock movement, racket direction, and dominant area representation using a 1–10 scale.
- The survey captures subjective feedback from badminton players to complement quantitative analysis.

Chapter 7

Conclusions and Future Works

Survey : Participants



- Participants (57 Responses):
  - Badminton club in Japan (Tachikawa)
  - Badminton club in Thailand (Bangkok)
  - Badminton club in Indonesia (Yogyakarta)

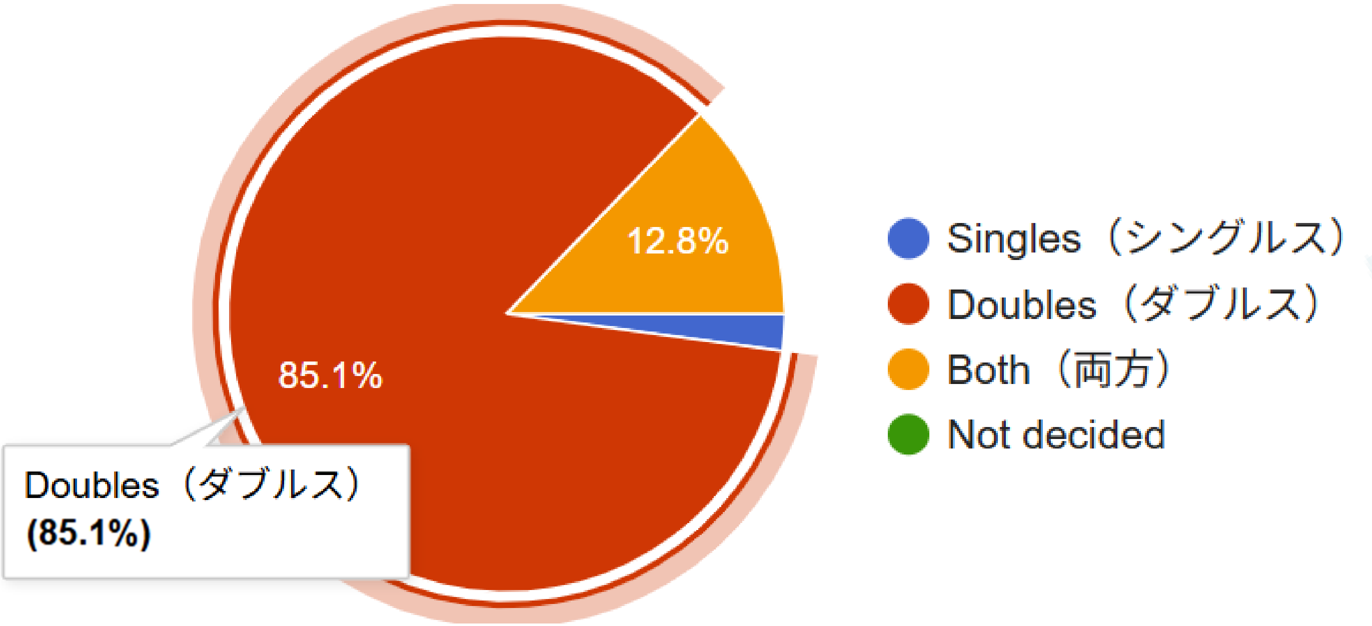


Figure 7.1. Distribution of respondents by duration of badminton playing experience.

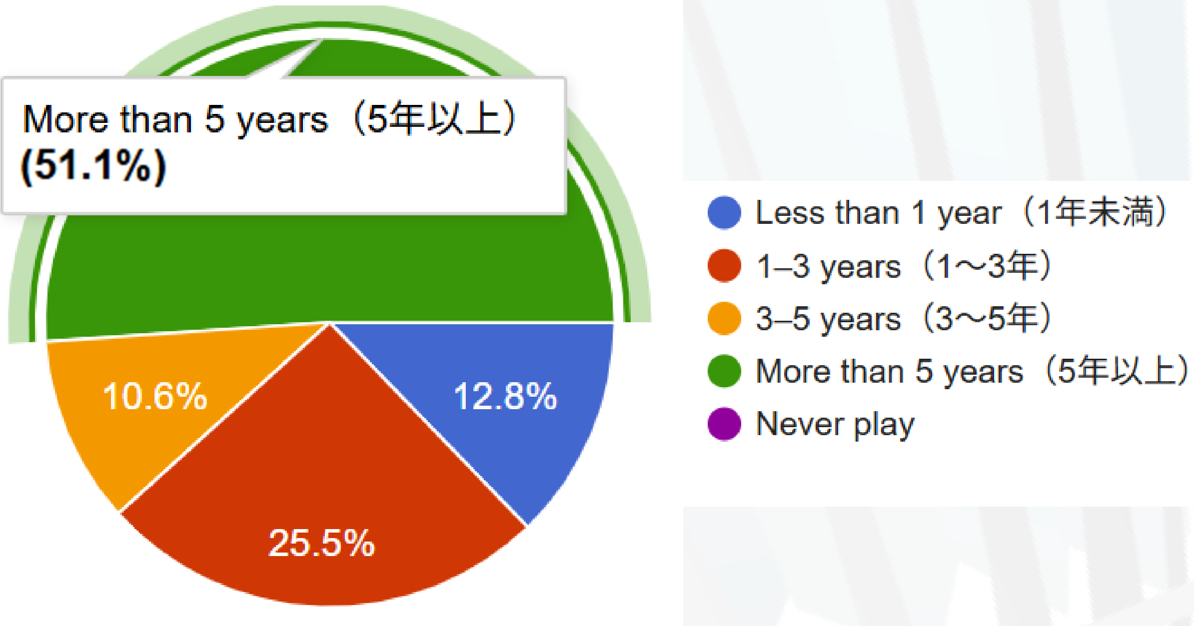


Figure 7.2. Distribution of respondents by usual match type played.

Conclusions and Future Works

Survey : User Evaluation Results

User evaluation scores were collected through a survey using a 1–10 rating scale. Player and Shuttlecock Detection receive consistently high scores, while Racket Detection shows greater variability and the Dominant Area is generally well evaluated.

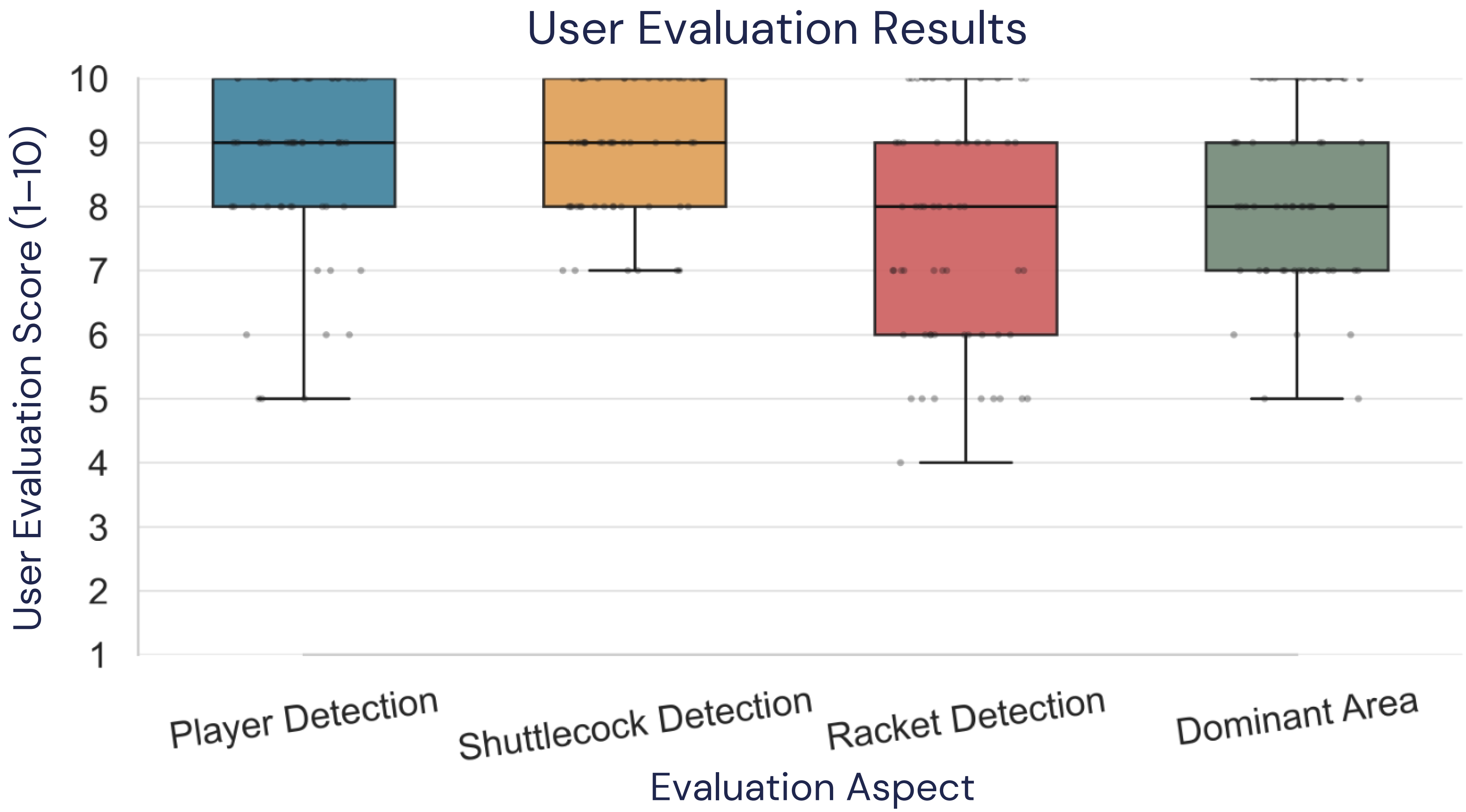


Figure 7.3. Distribution of user evaluation scores (1–10) across evaluation aspects.

Conclusions and Future Works

Conclusions

- This dissertation establishes a complete pipeline for badminton sports analysis, including player detection, shuttlecock trajectory estimation, racket detection, and dominant area estimation.

- Player detection was improved through heuristic YOLOv3 training, and tracking was implemented using SORT.

- Shuttlecock detection was enhanced by extending TrackNetV2 with residual learning to mitigate vanishing gradients, Leaky ReLU to avoid the dying ReLU problem, and post-residual batch normalization to prevent over-normalization.

Conclusions and Future Works

Conclusions (Cont.)

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- Racket detection was advanced by creating novel datasets: ORDB with oriented bounding boxes to explicitly model racket orientation in the detection results, detection was performed using YOLOv11-OBB and tracking was conducted with SORT.

- Together, these contributions deliver a Dominant area estimation integrated players, shuttlecock, and racket data, providing new insights into tactical regions in double matches. It aligns with sports science concepts of anticipation and court coverage.

- The user evaluation results show median scores of 9 for Player Detection and Shuttlecock Detection, and 8 for Racket Detection and the Dominant Area, indicating generally positive user perception across all aspects.

Conclusions and Future Works

Future Works

- Future work focuses on improving dominant area estimation by considering temporal factors such as reaction delay, and interactions between players to the shuttlecock natural movement, so that dominant area during fast rallies can be represented more realistically.

- The framework can be extended to analyze tactical behavior in doubles play, particularly by examining player roles in the front and back court and the coordination between teammates.

- In sport science perspective, future work can integrate fatigue, and player coordination dynamics.

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Publications

No	Event	Date	Type	Title
1	Institute of Image Information and Television Engineers (ITE) Conference 2025 – Yokohama, Japan	December 18, 2025	Local Conference	Integrating Multi-Object Information for Dominant Area Analysis in Badminton
2	ACM MMSports 2025 – ACM International Workshop on Multimedia Content Analysis in Sports - Dublin, Ireland	October 27, 2025	International Conference	Oriented Racket Database (ORDB): An Annotated Dataset for Rotated Racket Detection in Badminton
3	VISAPP 2025 – International Conference on Computer Vision Theory and Applications – Porto, Portugal	February 27, 2025	International Conference	RacketDB: A Comprehensive Dataset for Detection of Badminton Rackets
4	Institute of Image Information and Television Engineers (ITE) Conference 2024 – Tsukuba, Japan	March 21, 2024	Local Conference	Bunch of Tricks for Improving Shuttlecock Detection from Badminton Videos
5	IEECON 2024 International Electrical Engineering Congress - Bangkok, Thailand	March 7, 2024	International Conference	Bag of Tricks Toward Enhanced Shuttlecock Detection from Badminton Videos
6	International Journal on Informatics Visualization (JOIV)	December 15, 2023	Journal	Shuttlecock Detection Using Residual Learning in U-Net Architecture
7	ICIST 2023 International Conference on Information and Science Technology – Semarang, Indonesia	December 15, 2023	International Conference	Shuttlecock Detection Using Residual Learning in U-Net Architecture
8	Institute of Image Information and Television Engineers (ITE) Conference 2023 – Hokkaido, Japan	February 21, 2023	Local Conference	A Residual U-Net Architecture for Shuttlecock Detection
9	Institute of Image Information and Television Engineers (ITE) Conference 2022 – Tokyo, Japan	December 23, 2022	Local Conference	Badminton Player Positioning Comparison using Heatmap Data During BWF World Championship 2021
10	International Journal on Informatics Visualization (JOIV)	December 14, 2022	Journal	Improving Badminton Player Detection Using YOLOv3 with Different Training Heuristic
11	ICIST 2022 International Conference on Information and Science Technology – Semarang, Indonesia	December 14, 2022	International Conference	Improving Badminton Player Detection Using YOLOv3 with Different Training Heuristic
12	Institute of Image Information and Television Engineers (ITE) Conference 2022 – Fukushima, Japan	August 24, 2022	Local Conference	Badminton Player Detection and Heatmap Visualization Based on Convolutional Neural Networks
13	IES 2022 International Electronics Symposium – Surabaya, Indonesia	August 9, 2022	International Conference	Heatmap Visualization and Badminton Player Detection using Convolutional Neural Network

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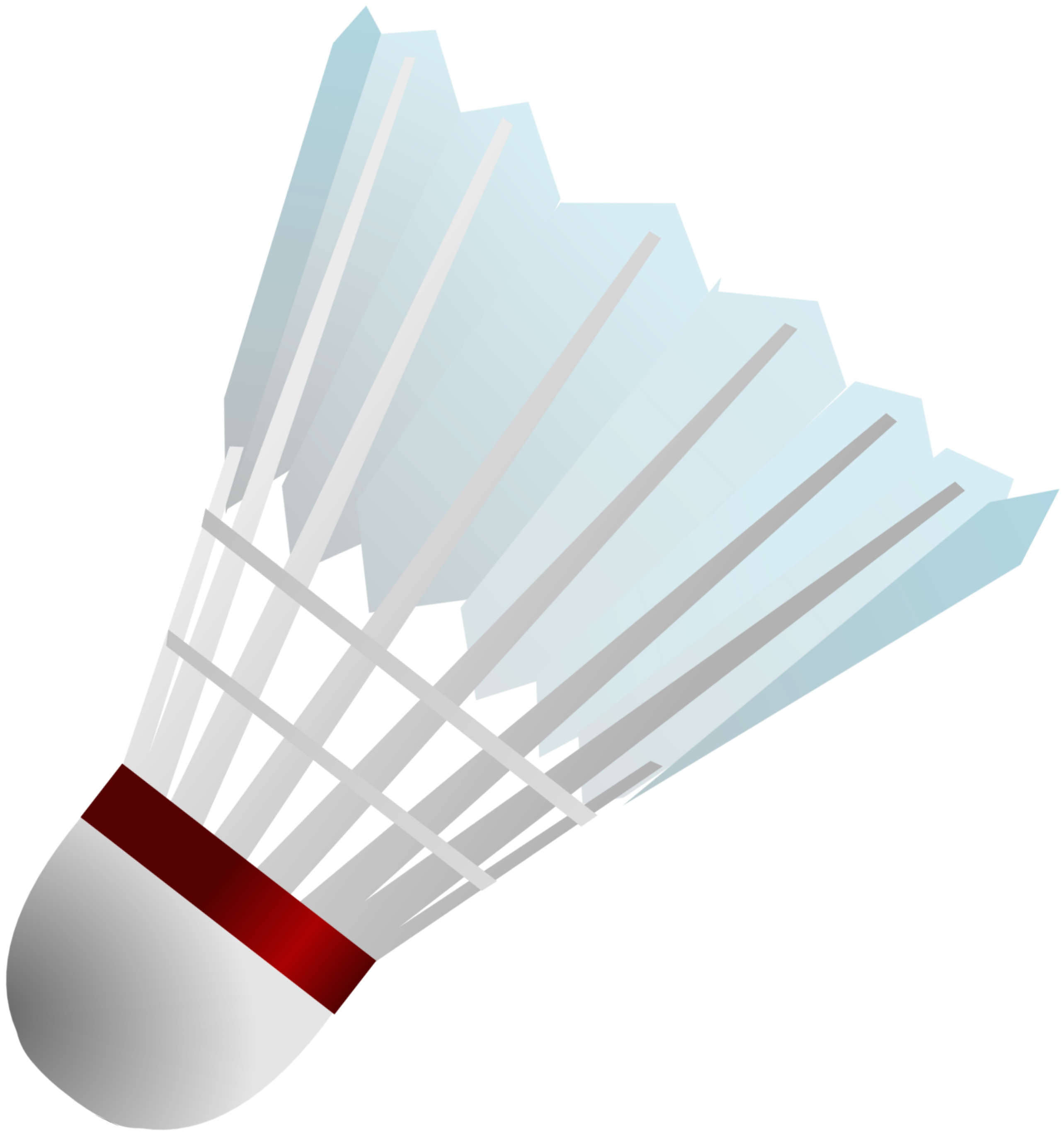

Thank you.

ありがとうございます